

The original kernel across the complete cutoff window Measured phase, heat, and full subquotients

Split-Zero research programme

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The original constrained kernel has a computed logarithmic volume at every source cutoff $N_j = q - 1 + j$, including all four prescribed endpoints:

$$\log \frac{\det H_{K,q-1}}{\det H_{K,N_j}} = mqJ(j/q) + o(kq), \quad 0 \leq j \leq q + 1.$$

Here $q = (k + 1)^2$, $m = 8k - 16$, the period and stipulated simple quartet are fixed, and the same complete kernel is retained throughout. Its actual relative spectra satisfy the stronger uniform logarithmic estimate IR6. The profile passes through every specified complete fixed subquotient of diverging rank.

The proof supplies uniform Gamma-power asymptotics through the first cutoff, the entire growing-jet minimum, and the original invariant-row comparison. An exact elliptic coordinate gives a convergent endpoint correction and certifies where half the limiting volume loss occurs. The original measured phase receives a moving transition; the measured heat retains its finite angle defect and receives a sharper physical smoothing error.

Both final incoming responses are incorporated. The separate complex arithmetic-current signs remain a further calculation. No RH proof or numerical evaluation of a hypothetical zero is claimed.

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The original cutoff and its measured return

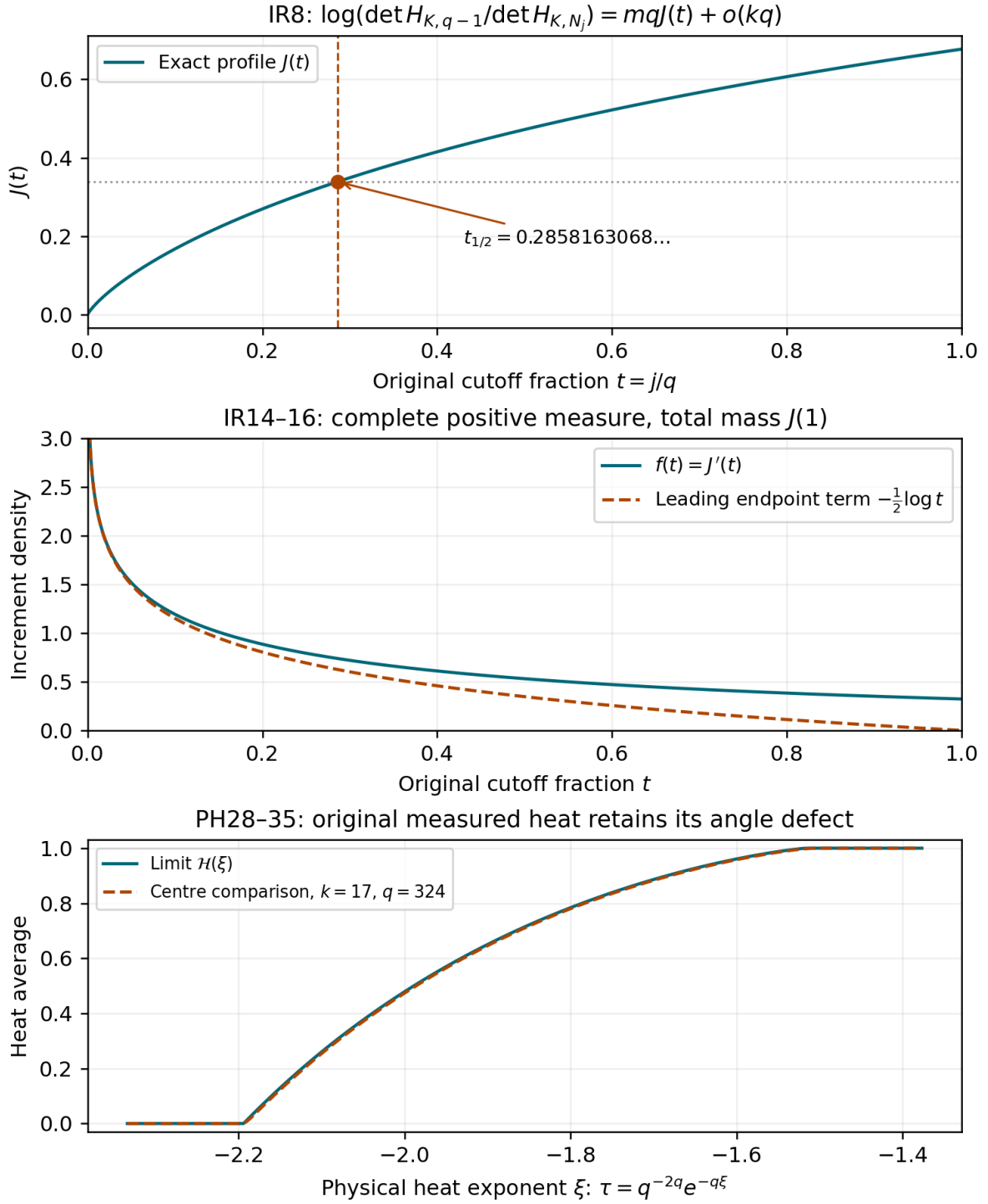


Figure 1: Top: the exact scalar in the original kernel-volume formula IR8. The half-volume position is certified in EP18. Middle: its increment density IR14-16, whose total mass is $J(1) = C_\partial/2$; the divergence at zero is integrable and the correction is proved in EP15-16. Bottom: the evaluated limiting heat distribution PH31 and its finite scalar-centre comparison. The dashed curve is not a numerical native response. PH34-35 bounds the actual response error and preserves its full angle defect. Elliptic formulas use B. C. Carlson's DLMF19.5 and 19.8, with original TeX and a full derivation in EP1-17. Plot sampling is numerical; the proofs and interval certificates are separate.

Part I

Original maps, complete subquotients, and receivers

The complete cutoff profile and its original receiving maps

Result **SZ-20260922-025** extends the evaluated original kernel coefficient to every cutoff between $q - 1$ and $2q$. The proof includes the full constrained kernel, each exceptional quotient, every affine source minimum and the separate positive-word and observed-heat maps. Its end-point returns agree with the published024 result; they are not added a second time.

IR1. Original objects, domain and complete result

Retain the original fixed hypothetical simple quartet with $0 < \delta < 1/2$, $\gamma > 2$, its admitted actual period, branch and arithmetic unit. Keep

$$k \geq 17, k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad q' = (k-7)^2, \quad m = 8k - 16, \quad \Delta = q - q' = 16k - 48. \quad (\text{IR1})$$

The coefficient space is $E_k = \mathbb{C}[y]/(Q_k)$, where

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta] = D_k(y)O_k(y). \quad (\text{IR2})$$

The physical coordinate is $S = k/2 + iy$, and the arithmetic action is $k/2 + iM$, $M[p] = [yp]$. The fixed original onto map $\Lambda : E_k \rightarrow B_k$ has its full kernel K_k and fixed frame I_K . This kernel includes every extra invariant row in addition to the conductor condition. Its rank is m . At the original complete source cutoff N , G_N is the attained coefficient metric and

$$H_{K,N} = I_K^* G_N I_K, \quad Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_{B,N}. \quad (\text{IR3})$$

All roots, difference factors, complex cross terms, Gamma mass $\sqrt{2\pi}$, and both real integration regions remain as in024. The original four-cutoff return is

$$\mathcal{R}F = F_{q-1} + F_q - F_{2q-1} - F_{2q}. \quad (\text{IR4})$$

Set $N_j = q - 1 + j$, $t_j = j/q$, $0 \leq j \leq q + 1$. Define J by EP1-EP8, equivalently the supplied CP3, with $J(0) = 0$. For

$$B_{ij} = H_{K,N_j}^{-1/2} H_{K,N_i} H_{K,N_j}^{-1/2}, \quad d_{ij} = q[J(t_j) - J(t_i)], \quad (\text{IR5})$$

the completed independent proof PP1-PP45 gives

$$\sup_{0 \leq i \leq j \leq q+1} \sum_{a=1}^m |\log \lambda_a(B_{ij}) - d_{ij}| = o(mq). \quad (\text{IR6})$$

Its finite estimate, before limits, is

$$\sum_a |\log \lambda_a(B_{ij}) - d_{ij}| \leq 2mq\eta_{k,\epsilon} + 2(v_0 + c_R)q[C_0 + J(1 + 1/q)]. \quad (\text{IR7})$$

Here $\eta_{k,\epsilon}$ is the full explicit quantity PP40; it includes the complete jet error PP36, the actual two-region source comparison and the arithmetic/Gamma width. The fixed residue error has codimension at most $v_0 + c_R$ *after restriction to the actual invariant kernel*. Its complete exceptional quotient has not been deleted. First $k \rightarrow \infty$ for fixed ϵ, R ; then $\epsilon \downarrow 0$. The proof does not exchange these limits or allow an unproved moving-period substitution.

Summing the actual relative eigenvalues proves

$$\log \frac{\det H_{K,q-1}}{\det H_{K,N_j}} = mqJ(j/q) + o(kq), \quad \text{uniformly for all } 0 \leq j \leq q + 1.$$

(IR8)

The original four signs give

$$\mathcal{R} \log \det H_{K,N} = mq[J(1) + J(1 + 1/q) - J(1/q)] + o(kq) = mC_\partial q + o(kq), \quad (\text{IR9})$$

because $J(1) = C_\partial/2$ and J is continuous. Both adjacent degrees are present. This recovers 024 CK14 without changing its coefficient.

IR2. Exact single-source growth and its limiting measure

Let ϕ_{N+1} be the next actual original source polynomial of unit norm, and $u_N = [\phi_{N+1}] \in E_k$ its entire remainder. The complete source covariance is the sum of its column outer products, so appending this source gives exactly

$$G_{N+1}^{-1} = G_N^{-1} + u_N u_N^*. \quad (\text{IR10})$$

In the actual metric at N , put

$$\rho_N = u_N^* G_N u_N, \quad b_N = I_K^* G_N u_N, \quad \chi_N = b_N^* H_{K,N}^{-1} b_N, \quad \beta_N = (\Lambda u_N)^* Q_{B,N} (\Lambda u_N). \quad (\text{IR11})$$

The two maps $P_K = I_K H_{K,N}^{-1} I_K^* G_N$ and $P_B = L_N \Lambda$ are complementary G_N -orthogonal projections. Indeed $\Lambda L_N = I$, $\Lambda I_K = 0$, $L_N^* G_N I_K = 0$, and their ranks sum to q . Thus $u_N = P_K u_N + P_B u_N$ gives

$$\rho_N = \chi_N + \beta_N. \quad (\text{IR12})$$

Multiplying $G_N - G_N u_N u_N^* G_N / (1 + \rho_N)$ by the covariance in IR10 gives the identity matrix directly. Restricting that inverse to the unchanged original kernel and taking its determinant gives

$$H_{K,N+1} = H_{K,N} - \frac{b_N b_N^*}{1 + \rho_N}, \quad a_{j,k} := \log \frac{\det H_{K,N_j}}{\det H_{K,N_{j+1}}} = \log \frac{1 + \rho_{N_j}}{1 + \beta_{N_j}} \geq 0. \quad (\text{IR13})$$

To verify the determinant step, conjugate by $H_{K,N}^{-1/2}$. The resulting rank-one perturbation has eigenvalue $1 - \chi_N / (1 + \rho_N) = (1 + \beta_N) / (1 + \rho_N) > 0$ on its one possible nontrivial direction and eigenvalue one on its orthogonal complement. This also covers $b_N = 0$. All complex pairings in IR11 are kept. The finite inverse update is the classical Sherman-Morrison identity; the direct proof here fixes its exact domain and convention rather than relying on its name.

Define the actual finite positive measure, with the last adjacent increment included,

$$\nu_k = \frac{1}{mq} \sum_{j=0}^q a_{j,k} \delta_{j/q}. \quad (\text{IR14})$$

Its mass tends to $J(1) = C_\partial/2$; it is not declared a probability. Put

$$\varepsilon_k = \max_{0 \leq j \leq q+1} \left| \frac{1}{mq} \log \frac{\det H_{K,N_0}}{\det H_{K,N_j}} - J(j/q) \right|.$$

IR6 gives $\varepsilon_k \rightarrow 0$, with its finite estimate IR7 divided by mq . For $0 \leq t \leq 1$, telescoping gives

$$|\nu_k([0, t]) - J(t)| \leq \varepsilon_k + \omega_J(1/q), \quad \max_j \frac{a_{j,k}}{mq} \leq 2\varepsilon_k + \omega_J(1/q). \quad (\text{IR15})$$

Here $\omega_J(h) = \sup |J(x) - J(y)|$ for $x, y \in [0, 1 + 1/q]$, $|x - y| \leq h$. To obtain the first bound use the exact index $\lfloor qt \rfloor + 1$; at $t = 1$ this is $q + 1$, not q . To obtain the second subtract the two endpoint determinant approximations. EP12 gives $\omega_J(h) \leq J(h)$.

For a continuously differentiable test φ , integration of

$$\varphi(s) = \varphi(1) - \int_s^1 \varphi'(t) dt$$

against the difference of the two finite measures, followed by Fubini, yields

$$\left| \int \varphi d\nu_k - \int_0^1 \varphi(t) f(t) dt \right| \leq \left(|\varphi(1)| + \int_0^1 |\varphi'(t)| dt \right) [\varepsilon_k + J(1/q)], \quad f(t) = \log \frac{1 + r_t}{1 - r_t}. \quad (\text{IR16})$$

The endpoint singularity of f is integrable by EP15. Piecewise linear approximation of any continuous test, together with the bounded total masses in IR15, proves $\nu_k \Rightarrow f(t) dt$. This is the

full first-response angular-increment conclusion. It does not differentiate a finite-grid error or infer an individual microscopic source allocation from a macroscopic profile.

For fixed $0 \leq a < b \leq 1$, let $i = \lfloor aq \rfloor$, $j = \lfloor bq \rfloor$. For all sufficiently large q , $i < j$. The original observable fractions obey the exact product identity

$$\left[\prod_{h=i}^{j-1} \frac{1 + \beta_{N_h}}{1 + \rho_{N_h}} \right]^{1/[m(j-i)]} = \exp \left(- \frac{\log \det H_{K,N_i} - \log \det H_{K,N_j}}{m(j-i)} \right) \longrightarrow \exp \left(- \frac{J(b) - J(a)}{b-a} \right). \quad (\text{IR17})$$

The exponent and its division by $m(j-i)$ are part of this explicitly defined scalar observable. IR13 proves its map from the unchanged source data.

IR3. Every complete fixed subquotient

Let $I_X : X \rightarrow K_k$ be a fixed coefficient injection and $R : X \rightarrow Z$ a fixed onto coefficient map, both independent of cutoff. Define

$$H_{X,N} = I_X^* H_{K,N} I_X, \quad Q_{Z,N} = (R H_{X,N}^{-1} R^*)^{-1}. \quad (\text{IR18})$$

The minimum section $H_{X,N}^{-1} R^* Q_{Z,N}$ maps z to a lift with norm squared $z^* Q_{Z,N} z$; every other lift differs by a vector of $\ker R$ orthogonal to it. Pythagoras proves that IR18 is the minimum over the entire affine fibre, with all cross terms retained.

For restriction, set $V_j = H_{K,N_j}^{1/2} I_X H_{X,N_j}^{-1/2}$. Direct multiplication gives

$$V_j^* V_j = I_X, \quad H_{X,N_j}^{-1/2} H_{X,N_i} H_{X,N_j}^{-1/2} = V_j^* B_{ij} V_j. \quad (\text{IR19})$$

For the complete quotient, put

$$C_{ij} = H_{X,N_i}^{1/2} H_{X,N_j}^{-1} H_{X,N_i}^{1/2}, \quad U_i = Q_{Z,N_i}^{1/2} R H_{X,N_i}^{-1/2}.$$

Then

$$U_i U_i^* = I_Z, \quad Q_{Z,N_i}^{1/2} Q_{Z,N_j}^{-1} Q_{Z,N_i}^{1/2} = U_i C_{ij} U_i^*. \quad (\text{IR20})$$

The eigenvalues of C_{ij} equal those of the relative pair in IR19: they are the eigenvalues of the two matrices AA^* and A^*A , for $A = H_{X,N_j}^{-1/2} H_{X,N_i}^{1/2}$, up to exchanging their order. The corresponding equality holds for the two orientations of the relative quotient matrix in IR20. These are similarities of positive matrices, not identifications of their eigenvectors.

To see precisely why the bounded exceptional count persists, suppose a self-adjoint matrix has at most c eigenvalues above u . Its quadratic form is at most u on the orthogonal complement of their spectral space. If an isometric compression had $c+1$ eigenvalues above u , its corresponding image would intersect that complement nontrivially, a contradiction. The argument below a lower threshold is identical with the inequality reversed. IR19 and IR20 therefore preserve PP42–PP44's at most c exceptions at each end, and preserve its full coarse interval $1 \leq \lambda \leq e^{C_0 q}$.

For $d_k = \dim Z \rightarrow \infty$, the identical finite argument gives

$$\sum_{a=1}^{d_k} \left| \log \lambda_a(Q_{Z,N_j}^{-1/2} Q_{Z,N_i} Q_{Z,N_j}^{-1/2}) - d_{ij} \right| \leq 2d_k q \eta_{k,\epsilon} + 2(v_0 + c_R)q[C_0 + J(1 + 1/q)]. \quad (\text{IR21})$$

First let $k \rightarrow \infty$, then $\epsilon \downarrow 0$, as before. Dividing by $d_k q$ proves CQ3, uniformly over all pairs of cutoffs. Determinants and the entire increment measure IR14–IR16 follow with m replaced by d_k . No lower bound on the growth rate of d_k is needed beyond its divergence: the exceptional codimension is fixed before this limit. For fixed d_k , IR21 remains a finite bound; its exceptional term cannot be dropped. This retains rather than assumes the missing information in that different rank regime.

IR4. Measured phase and heat receivers

Keep $T = D_k(M)$, $H_N = G_N^{-1} T^* G_N T$ and the same L_N, Λ . The full proofs PH1–PH41 supply the exact image map, all-vector word metric, every compressed kernel energy and the resolvent determinant. In particular, with

$$c_k(t) = 2q \log q + q[2 \log(4/\pi) - 2 - J(t)],$$

the complete positive word spectrum has logarithms $c_k(t_j) + o(q)$ uniformly over every direction and cutoff. The paired compressed-kernel energies have total logarithmic transport $o(kq)$. PH18–PH20 then prove for every cutoff pair

$$\frac{2}{\pi} \int_0^\infty (\Theta_{N_i}(\omega) - \Theta_{N_j}(\omega)) \frac{d\omega}{\omega} = \Delta q[J(t_j) - J(t_i)] + o(kq). \quad (\text{IR22})$$

The finite error is $\mathcal{B}_{ij} = D_{ij} + E_i^W + E_j^W$ in PH15, and the full $L^1(d\omega/\omega)$ error is at most $\pi \mathcal{B}_{ij}/2$. Both frequency ends are included. Adding pairs $(0, q)$ and $(1, q+1)$ gives $+\Delta C_\partial q + o(kq)$, preserving the original four signs. At $\omega = q^{2q} e^{bq}$, the pair phase divided by k approaches height 8π between the evaluated boundaries $\eta(t_j)$ and $\eta(t_i)$; the original four-cutoff height is 16π .

The first response's separate heat calculation is retained in PH22–PH41. It uses the physical time $\tau = q^{-2q} e^{-q\xi}$ and the q declared averaging cutoffs $j = 0, \dots, q-1$. The original integer-subtracted measured heat M_k has the exact decomposition $M_k = F_k + D_k^{\text{ang}}$, where F_k is the full positive-word heat average and $0 \leq D_k^{\text{ang}} \leq \alpha_k$. Its left endpoint is α_k , not zero. PH34 sharpens its uniform smoothing error to

$$\varepsilon_k^{\text{heat}} = \frac{\delta_k}{f(1)} + \frac{1}{q} + \frac{\gamma_E + 2E_1(1)}{qf(1)}. \quad (\text{IR23})$$

The actual measured error is bounded by this quantity plus its retained angle defect. The affine map to a probability heat is proved in PH36, and its complete transport and exact Euler-constant bias are in PH38–PH40. These are additional observables connected to the original one, not replacements for it.

IR5. Corrections, checks, and the continuing programme

Both final responses were already in the incoming paste. Comparison with the full session transcript found only formatting, source-link display, and duplicated attachment labels. Their distinct mathematical passages are both included: first-response heat, rank-one increments and endpoint expansion; second-response complete subquotients.

The exact scalar computation keeps a term that the incoming text's phrase about cancelling every power of n omitted: PP24 contains $-\frac{1}{2} \log n$ in its finite remainder. The leading coefficient is unchanged. EP16 strengthens the endpoint expansion and EP18 independently certifies the half-volume position. Incoming scripts were not provided with the files; their reported 59 exact checks are recorded as supplied reports. This edition's independent exact and interval checks have their own receipts and scope.

This result extends control of the original kernel, word energies, observed resolvent and measured heat throughout the source window. The separate complex arithmetic current remains the original insertion

$$i(Q_{B,N} M_{B,N} - M_{B,N}^* Q_{B,N}). \quad (\text{IR24})$$

Its individual marked-class pairings involve the actual complex response ratios and complete-action cancellation. The positive-word spectra in IR22 do not assign their signs. The exact original current and response maps remain in the retained proof sources; the next programme calculation is their phase-bearing numerator. No result here claims an RH proof or a hypothetical zero computation.

Part II

Uniform Gamma powers and every original jet

The Gamma-power profile at every original cutoff

Independent derivation for the 025 continuation. The unchanged finite Gamma estimates from 024 PJ1-PJ44 and the original-source return CK5-CK11 are retained. Their source mass, root-gap constants, arithmetic data, and complete minima are not altered. This proof extends their parameter range and supplies the moving-potential argument needed for the new uniform cutoff conclusion.

PP1. Original Gamma objects and exact parity

Keep

$$d\sigma(y) = w(y)dy = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \sigma(\mathbb{R}) = \sqrt{2\pi}, \quad d\nu_n = |y|^{2n} d\sigma, \quad d_n = \int d\nu_n. \quad (\text{PP1})$$

Let $U_j^{[n]}$ be the monic orthogonal polynomial for ν_n , let $h_j^{[n]} = \|U_j^{[n]}\|_{\nu_n}^2$, and let $K_{n,M}$ be the complete polynomial reproducing kernel through degree M . Define the actual scalar

$$F_n(M) = \log[d_n K_{n,M}(0, 0)]. \quad (\text{PP2})$$

Since the constant polynomial has norm squared d_n , $F_n(0) = 0$. Positivity of the kernel sum makes $F_n(M)$ nonnegative and nondecreasing in M .

For $n \geq 36$, the accepted PJ16 and PJ18, including the last raising degree, give fixed $0 < c \leq 1 \leq C$ such that

$$cn\|P\|_{\nu_n} \leq \|yP\|_{\nu_n} \leq Cn\|P\|_{\nu_n} \quad (\deg P \leq 2n+1), \quad (\text{PP3})$$

and all nonzero roots of the used $U_j^{[n]}$, $j \leq 2n+2$, lie in $[-Cn, -cn] \cup [cn, Cn]$. The odd zero at zero is simple and is retained separately. The constants arise from the complete two-half-line Laguerre comparison and its near-zero correction in PJ6-PJ16; no pointwise lower Gamma comparison at zero is used.

For even $M = 2h$, the monic Christoffel-Darboux identity gives

$$F_n(M) = \log d_n + 2 \log |U_M^{[n]}(0)| - \log h_M^{[n]} + \log D_{n,M}, \quad D_{n,M} = \frac{(U_{M+1}^{[n]})'(0)}{U_M^{[n]}(0)}. \quad (\text{PP4})$$

For $h \geq 1$, the positive roots interlace as $0 < a_1 < b_1 < a_2 < b_2 < \dots < a_h < b_h$. Hence

$$1 \leq D_{n,M} = \prod_{i=1}^h \frac{b_i^2}{a_i^2} \leq \frac{b_h^2}{a_1^2} \leq (C/c)^2. \quad (\text{PP5})$$

The middle inequality uses $b_i < a_{i+1}$, so it telescopes. For $M = 0$, $U_0 = 1$, $U_1 = y$, and $D_{n,0} = 1$ exactly. For odd M , parity gives the exact same-measure identity

$$F_n(M) = F_n(M-1). \quad (\text{PP6})$$

No change of the power n is needed for this parity step.

PP2. Uniform monic norms at positive degree/power ratios

Fix a compact interval $I = [a_*, b_*] \subset (0, 3/2]$. Here a_*, b_* are ratio bounds, not equilibrium endpoints. Let $M = 2h$ and $\tau = M/n \in I$. The exact coordinate map is

$$U_M^{[n]}(y) = n^M p_h(y^2/n^2), \quad p_h \text{ monic of degree } h. \quad (\text{PP7})$$

Reflection and uniqueness give the parity of the original minimizer, and substituting every even test monomial shows that p_h is the monic orthogonal polynomial for the full pushforward

$$d\eta_n(x) = n^{2n+1} x^{n-1/2} w(n\sqrt{x}) dx, \quad x > 0. \quad (\text{PP8})$$

Both real half-lines contribute to this formula. Consequently

$$h_M^{[n]} = n^{2(n+M)+1} \min_{\substack{p \text{ monic} \\ \deg p=h}} \int_0^\infty x^{n-1/2} |p(x)|^2 w(n\sqrt{x}) dx. \quad (\text{PP9})$$

The varying potential is exactly

$$V_\tau(x) = \frac{\pi}{\tau} \sqrt{x} - \frac{2}{\tau} \log x, \quad hV_\tau(x) = \frac{\pi n}{2} \sqrt{x} - n \log x. \quad (\text{PP10})$$

Use the already proved EIQ family at $(\alpha, \beta) = (2/\tau, \pi/\tau)$. Its probability μ_τ has support $[u_\tau^2, v_\tau^2]$, with $u_\tau K(\kappa_\tau) = 2$, $v_\tau E(\kappa_\tau) = 2(1 + \tau)$, and $\kappa_\tau^2 = 1 - u_\tau^2/v_\tau^2$. Its Euler constant and logarithmic moment are denoted by λ_τ and L_τ . In particular,

$$V_\tau(x) - 2 \int \log |x - z| d\mu_\tau(z) = \lambda_\tau \text{ on its support, } \geq \lambda_\tau \text{ on } (0, \infty). \quad (\text{PP11})$$

The parameter continuity proved in EIQ29 puts all these supports inside one compact interval separated from zero, with lengths bounded below. The positive prefactor in EIQ10–EIQ11 is continuous and bounded above and below there. Thus their densities are uniformly bounded and their square-root endpoint entropy integrals are uniformly finite. The constants λ_τ are bounded on I .

We spell out why the monic argument is uniform rather than applying only its diagonal conclusion. On the support, Jensen against its arcsine probability bounds every monic degree- h polynomial below by $c_I e^{-h\lambda_\tau}$ in the weight $e^{-hV_\tau} x^{-3/4} dx$. Indeed the logarithmic potential of that arcsine probability is its log capacity on the interval and no smaller at any complex root; averaging (PP11) against it gives the exact Euler constant. The entropy integral of $x^{-3/4}$ against the arcsine density is bounded below uniformly because the endpoints stay separated from zero and their distance stays bounded above and below. This lower bound uses only the equilibrium interval.

For the upper bound, choose the h midpoint quantiles of μ_τ as zeros. Their distribution functions have discrepancy at most $1/(2h)$. On any fixed bounded x -interval containing the supports, truncation of $\log |x - z|$ at distance $1/h$ has variation $O_I(\log(h+2))$, and the bounded equilibrium density makes its integral truncation error $O_I(1/h)$. The integration-by-parts formula for a function of bounded variation against the distribution-function difference therefore gives

$$\log |p_h(x)| \leq h \int \log |x - z| d\mu_\tau(z) + C_I \log(h+2).$$

Using (PP11), its full bounded-region integral is at most $C_I(h+2)^{C_I} e^{-h\lambda_\tau}$, because $x^{-3/4}$ is integrable at zero. For a sufficiently large common B , all roots are in $(0, B/2)$, and $|p_h(x)| \leq x^h$ for $x \geq B$. Uniformly over $\tau \in I$,

$$(2 + 2/\tau) \log x - (\pi/\tau) \sqrt{x} \leq -\lambda_\tau - c_I \sqrt{x} \quad (x \geq B).$$

This follows from $\log x/\sqrt{x} \rightarrow 0$, the bounded λ_τ , and the positive lower bound for π/τ . It bounds the remaining tail integral by a fixed multiple of $e^{-h\lambda_\tau}$. Thus both ends of the actual integration domain are included.

PJ13 supplies fixed constants $A, B_0 > 0$ such that

$$w(y) \leq B_0 |y|^{-1/2} e^{-\pi|y|/2} \quad (y \neq 0), \quad w(y) \geq A |y|^{-1/2} e^{-\pi|y|/2} \quad (|y| \geq 1). \quad (\text{PP12})$$

The lower bound applies on every equilibrium interval once n is large, uniformly over I . Substituting (PP12) in (PP9), the full scale factor is $n^{2(n+M)+1/2}$, and the remaining weight is exactly $e^{-hV_\tau} x^{-3/4} dx$. Applying the interval lower bound and the global upper trial just proved gives

$$\log h_M^{[n]} = [2(n+M) + \tfrac{1}{2}] \log n - \frac{M}{2} \lambda_\tau + O_I(\log(n+2)). \quad (\text{PP13})$$

This proves CP17 on its stated compact positive ratio range. It is not applied to $M/n \rightarrow 0$. The original mass calculation, unchanged from PJ49, gives

$$\log d_n = 2n \log n + a_d n + r_d(n), \quad a_d = 2 \log(4/\pi) - 2, \quad r_d(n) = O(\log(n+2)). \quad (\text{PP14})$$

PP3. Moving-potential zero statistics on the unbounded domain

Let $n \rightarrow \infty$, $M = 2h$, and $\tau = M/n \rightarrow t \in (0, 3/2]$. By PP1 the zeros of p_h from (PP7) lie in the fixed interval $[c^2, C^2]$. This uses the actual degree $M+1 \leq 3n/2+1 < 2n+2$ in the original root-gap bound.

For a positive measure ζ , let $D_h(\zeta)$ be its monomial Gram determinant through degree $h-1$. The exact identities from PJ62 and PJ67 are

$$D_h(\zeta) = \frac{1}{h!} \int \prod_{i < j} (x_i - x_j)^2 \prod_i d\zeta(x_i), \quad (\text{PP15})$$

$$(-1)^h p_h(-A) = \frac{D_h((A+x)\eta_n)}{D_h(\eta_n)} \quad (A > 0). \quad (\text{PP16})$$

They follow by expansion of the two Vandermonde determinants and of the monic moment determinant; in particular they retain the full measure factor. Set

$$\phi(x) = \log(A+x), \quad f_n(z) = \log D_h((A+x)^z \eta_n), \quad \kappa_n = n^{2n+1/2},$$

$$\mathcal{F}_t(\chi) = \sup_{\pi} \left\{ \iint \log |x-y| d\pi^2 - \int V_t d\pi + \chi \int \phi d\pi \right\}. \quad (\text{PP17})$$

The supremum is defined by its bounded-above energy kernel on probabilities on $(0, \infty)$. We prove for the actual moving sequence

$$\limsup \frac{f_n(h\chi) - h \log \kappa_n}{h^2} \leq \mathcal{F}_t(\chi), \quad \liminf \frac{f_n(0) - h \log \kappa_n}{h^2} \geq \mathcal{F}_t(0). \quad (\text{PP18})$$

For the upper bound, (PP12) gives globally $d\eta_n \leq B_0 \kappa_n e^{-hV_\tau(x)} x^{-3/4} dx$. Use the probability $d\omega_0 = (2\sqrt{\pi})^{-1} x^{-3/4} e^{-\sqrt{x}} dx$. Put $Q_{\tau,\chi} = V_\tau - \chi\phi$ and $K_{\tau,\chi}(x,y) = \log |x-y| - [Q_{\tau,\chi}(x) + Q_{\tau,\chi}(y)]/2$. The exponent in (PP15) is exactly

$$\sum_{i \neq j} K_{\tau,\chi}(x_i, x_j) + \sum_i [\sqrt{x_i} - Q_{\tau,\chi}(x_i)]. \quad (\text{PP19})$$

The latter summand is bounded above uniformly for τ in a compact positive neighborhood of t within $(0, 3/2]$, and χ in a fixed small interval. Its square-root coefficient at infinity is at most $1 - 2\pi/3 < 0$; at zero the logarithmic barrier has a uniformly positive coefficient. The factorial and constant prefactors cost only $O(h \log(h+2))$.

Here is an explicit bound for the moving potential before compactification. Let $H(x) = 1 + \sqrt{x} + |\log x|$. Then

$$|V_\tau(x) - V_t(x)| \leq e_n H(x), \quad e_n = (\pi+2)|1/\tau - 1/t| \rightarrow 0. \quad (\text{PP20})$$

For a fixed small $e > 0$, eventually $K_{\tau,\chi} \leq K_{t,\chi}^{(e)}$, where $K_{t,\chi}^{(e)} = K_{t,\chi} + e[H(x) + H(y)]/2$. Take e sufficiently small that this kernel still has strictly positive barriers at zero and infinity. Its truncation $\max(K_{t,\chi}^{(e)}, -T)$ is continuous on the compact square obtained by adding zero and infinity. At each boundary and along the diagonal the untruncated kernel tends uniformly to minus infinity. For the empirical probability L_h , omission of the diagonal therefore costs only T/h :

$$h^{-2} \sum_{i \neq j} K_{\tau,\chi}(x_i, x_j) \leq \iint \max(K_{t,\chi}^{(e)}, -T) dL_h^2 + T/h.$$

Integrating against ω_0^h bounds its exponential by the maximum of this continuous energy. Let first $n \rightarrow \infty$, then $T \rightarrow \infty$, then $e \downarrow 0$. These maxima decrease to $\mathcal{F}_t(\chi)$. To justify both passages, choose maximizing probabilities on the compactification, take a weakly convergent subsequence, compare with every fixed continuous truncation, and only then let the truncation increase. The positive barriers are uniform for all the chosen e , so a limiting boundary mass has energy minus infinity and cannot improve the finite test energy at μ_t . The kernels decrease pointwise when $e \downarrow 0$. This proves the first bound in (PP18) and gives the missing unbounded-domain justification for moving τ .

For the lower bound restrict every variable to the fixed support of μ_t , rather than moving the integration domain. The Gamma lower envelope holds there for all large n . On this compact interval $V_\tau - V_t \rightarrow 0$ uniformly by (PP20). Change the measure in (PP15) to $\mu_t^{\otimes h}$ and apply Jensen. The exponent is

$$h \log \kappa_n + h(h-1) \iint \log |x-y| d\mu_t^2 - h^2 \int V_t d\mu_t + o(h^2) + O(h) - \log(h!).$$

The entropy integral is finite because the density has the proved square-root endpoints with a positive bounded factor, and this interval is separated from zero. EIQ uniqueness identifies its energy as $\mathcal{F}_t(0)$. This proves the second bound in (PP18); the upper bound at zero gives a limit there.

For completeness, the differentiability used next is a property of the variational value, not of an asymptotic remainder. Uniformly for small χ , $K_{t,\chi} \leq C_0 - c_0[H(x) + H(y)]$ after adjusting the fixed constant. Maximizers exist by the same compactification argument, and testing on μ_t bounds their H -moments. Their ϕ -integrals are uniformly integrable because ϕ is bounded at zero and $\phi(x)/\sqrt{x} \rightarrow 0$ at infinity. Every limit of maximizers as $\chi \rightarrow 0$ maximizes the unperturbed energy, by its upper semicontinuity obtained through fixed truncated kernels. EIQ uniqueness forces that limit to be μ_t . Testing the two potentials on each other's maximizers bounds their secant by the corresponding two ϕ -integrals. Hence

$$\mathcal{F}'_t(0) = \int \log(A+x) d\mu_t(x). \quad (\text{PP21})$$

Hölder's inequality in (PP15) makes f_n convex. For fixed $\epsilon > 0$ and $h\epsilon \geq 1$,

$$\frac{f_n(0) - f_n(-h\epsilon)}{h^2\epsilon} \leq \frac{f_n(1) - f_n(0)}{h} \leq \frac{f_n(h\epsilon) - f_n(0)}{h^2\epsilon}. \quad (\text{PP22})$$

The exact term $h \log \kappa_n$ cancels. Apply (PP18) in its two stated directions, let $n \rightarrow \infty$, and only then $\epsilon \downarrow 0$. Equations (PP16) and (PP21) show that the zero probability of p_h has limiting $\log(A+x)$ -integrals equal to those of μ_t , for every $A > 0$. All its zeros lie in $[c^2, C^2]$. Every subsequential weak limit is therefore compactly supported. Expansion of $\log(A+x)$ in powers of $1/A$, for A larger than both supports, identifies every moment; polynomial approximation identifies the measures. Thus the zero probabilities converge to μ_t .

The original root gap makes $\log x$ a bounded continuous test on their common compact support separated from zero. Since $\mu_\tau \rightarrow \mu_t$ in this same regime, its logarithmic moments satisfy $L_\tau \rightarrow L_t$. Restoring the exact factor n^M in (PP7) gives

$$\log |U_M^{[n]}(0)| = M \log n + \frac{M}{2} L_\tau + r_U(n, M), \quad r_U(n, M) = o(n). \quad (\text{PP23})$$

This holds along every sequence with $M/n \rightarrow t > 0$ in the stated range. If the error were not uniform on a fixed compact positive ratio interval, choose a sequence violating uniformity, then take a convergent subsequence of its ratios. The just-proved sequential result contradicts the violation. This proves precisely the compact-positive uniformity claimed in CP22.

PP4. The two-parameter scalar, with its exact finite remainder

For even M , write the remainder in (PP13) as

$$r_h(n, M) = \log h_M^{[n]} - [2(n+M) + \frac{1}{2}] \log n + \frac{M}{2} \lambda_\tau.$$

Then $r_h = O_I(\log(n+2))$. Substitution of (PP13), (PP14), and (PP23) into (PP4) gives the exact identity

$$F_n(M) - n\Phi(\tau) = r_d(n) - r_h(n, M) + 2r_U(n, M) - \frac{1}{2} \log n + \log D_{n,M}, \quad (\text{PP24})$$

$$\Phi(t) = 2 \log(4/\pi) - 2 + tL_t + \frac{t}{2} \lambda_t. \quad (\text{PP25})$$

The term $-\frac{1}{2} \log n$ is retained here. The leading $n \log n$ terms cancel; saying that every power of n cancels exactly would discard this finite term. It is part of the allowed $O(\log n) + o(n)$ error, so it changes none of the proposed limits.

The general EL3-EL4 formulas, read in the retained024 proof, give the exact dictionary

$$\Phi(t) = 2 \log(4/\pi) + 2(1+t) \log w_t - 2 \log z_t - t \log c_t,$$

where $w_t = (u_t + v_t)/2$, $z_t = u_t v_t$, and $c_t = (v_t^2 - u_t^2)/4$. Substituting the actual $v_t = 2(1+t)/E(\kappa_t)$, $u_t = r_t v_t$, gives

$$\Phi(t) = 2 \log \frac{E(\kappa_t)(1+r_t)^{1+t}}{\pi(1+t)r_t \kappa_t^t} = \mathfrak{J}(t). \quad (\text{PP26})$$

No arithmetic kernel coefficient enters this equality. The original elliptic equations make $\mathfrak{J}$ continuous on $(0, 3/2]$. At zero, the retained small-modulus expansions give $t = \kappa_t^4/16 + O(\kappa_t^6)$, $r_t \rightarrow 1$, and $t \log \kappa_t \rightarrow 0$; hence the displayed quotient tends to one and $\mathfrak{J}(t) \rightarrow 0$. Set $\mathfrak{J}(0) = 0$.

Equations (PP5) and (PP24) prove uniform convergence on compact positive ratios for even degrees. For odd degrees use (PP6) and uniform continuity of $\mathfrak{J}$; the argument is always in the same ν_n . To prove uniformity at residual degree zero, fix $0 < d < 3/2$. For every $M \leq dn$, monotonicity gives

$$0 \leq F_n(M)/n \leq F_n(\lceil dn \rceil)/n \longrightarrow \mathfrak{J}(d). \quad (\text{PP27})$$

This also proves nonnegativity and monotonicity of $\mathfrak{J}$ by taking fixed positive degree ratios and passing to the limit. Therefore $0 \leq \mathfrak{J}(M/n) \leq \mathfrak{J}(d)$, and

$$\limsup_n \sup_{M \leq dn} |F_n(M)/n - \mathfrak{J}(M/n)| \leq \mathfrak{J}(d).$$

Let $d \downarrow 0$. Combining this with compact-positive convergence proves

$$\boxed{\epsilon_n^G := \max_{0 \leq M \leq \lfloor 3n/2 \rfloor} |F_n(M)/n - \mathfrak{J}(M/n)| \longrightarrow 0.} \quad (\text{PP28})$$

This proves CP9, including $M = 0$, every finite odd degree, and every sequence with $M/n \rightarrow 0$. CP17 is only used where its compact-positive hypothesis holds.

The exact finite matrix for this scalar is also fixed. Put $h = \lfloor M/2 \rfloor$ and

$$\mathcal{A}_{ab} = \frac{d_{n+a+b}}{n^{2(a+b)}}, \quad 0 \leq a, b \leq h. \quad (\text{PP29})$$

This is the full even coefficient Gram in the variable y^2/n^2 and the literal ν_n . The odd block is orthogonal to it. Evaluation at zero gives $K_{n,M}(0,0) = (\mathcal{A}^{-1})_{00}$, so $e^{F_n(M)} = d_n(\mathcal{A}^{-1})_{00}$. Writing the same number as $((\mathcal{A}/d_n)^{-1})_{00}$ is an exact matrix identity, not a change to the physical source measure. This is the universally computable scalar remainder in (PP28).

PP5. Complete growing jets at every one of the original cutoffs

Retain $q = (k+1)^2 \geq 40$, $1 \leq s \leq q/10$, $n = q - s$, $N_j = q - 1 + j$, $0 \leq j \leq q + 1$. The exact jet map is

$$J_{q,s}P = (q^a[y^a]P)_{0 \leq a < s}.$$

Multiplication by y^{q-s} is the original norm-preserving map from the ν_n problem to the power-quotient fibre. Its permitted polynomial degree is exactly

$$M_j = N_j - (q - s) = s + j - 1. \quad (\text{PP30})$$

Let H_j^{pow} be its attained jet metric. The complete low coefficient estimate PJ19–PJ25 gives

$$d_n e^{-E(q,s)} I \leq H_0^{\text{pow}} \preceq d_n e^{E(q,s)} I, \quad E(q,s) = C_*[s \log(C_* q/s) + s + \log(q+2)]. \quad (\text{PP31})$$

For $j \geq 1$, the coefficient covariance is the full matrix $J_{q,s} J_{q,s}^* = (H_j^{\text{pow}})^{-1}$. The upper bound in PJ28–PJ30 uses the actual root products of every even and odd orthogonal polynomial through degree M_j , and the complete derivative-evaluation norm on the odd sector. Since $M_j \leq q + s < 3n/2$, the same binomial bounds and the same $E(q,s)$ work simultaneously for every j . They yield

$$(H_j^{\text{pow}})^{-1} \preceq e^{E(q,s)} K_{n,s+j-1}(0,0) I.$$

For the opposite inequality use the actual polynomial

$$H(y) = K_{n,j-1}(y,0)/K_{n,j-1}(0,0), \quad \mathcal{S}_j a(y) = H(y) \left[\text{rem}_{x^s} \frac{a(x)}{H(qx)} \right]_{x=y/q}. \quad (\text{PP32})$$

Its constant value is $H(0) = 1$, and its nonzero roots obey the same gap. Its squared norm is exactly $1/K_{n,j-1}(0,0)$. The reciprocal coefficients in (PP32) are bounded by the complete homogeneous products of inverse squared roots; their sum through degree $s-1$ is at most $\exp(E(q,s)/C)$ after enlarging the fixed constant. The truncated series multiplies back to the entire requested jet a , so $J_{q,s} \mathcal{S}_j = I$ and $\deg \mathcal{S}_j a \leq (j-1) + (s-1) = M_j - 1$. Repeated application of (PP3), with each original factor $1/q$, controls every increasing degree, including the last raising step. Its highest degree is below $M_j \leq q + s < 3n/2 < 2n + 2$. Thus $\|\mathcal{S}_j a\|_{\nu_n}^2 \leq e^{E(q,s)} \|a\|_2^2 / K_{n,j-1}(0,0)$. Taking the complete minimum proves

$$\boxed{e^{-E(q,s)} K_{n,j-1}(0,0) I \preceq (H_j^{\text{pow}})^{-1} \preceq e^{E(q,s)} K_{n,s+j-1}(0,0) I, \quad 1 \leq j \leq q+1.} \quad (\text{PP33})$$

For $j = 1$, the kernel polynomial in (PP32) is exactly $H = 1$; its reciprocal estimate and right inverse remain valid. For $j = 0$, use (PP31) and no negative-index kernel. This proves the entire degree range of CP26.

Combining (PP31) and (PP33) gives for every nonzero coefficient vector

$$F_n(j-1) - 2E(q, s) \leq \log \frac{a^* H_0^{\text{pow}} a}{a^* H_j^{\text{pow}} a} \leq F_n(s+j-1) + 2E(q, s). \quad (\text{PP34})$$

All matrices are complete attained minima, not diagonal row estimates.

Define on $[0, 3/2]$ the exact modulus of continuity $\omega_{\mathfrak{J}}(h) = \sup_{|u-v| \leq h} |\mathfrak{J}(u) - \mathfrak{J}(v)|$, and put $J_* = \max_{[0, 3/2]} \mathfrak{J}$. For $1 \leq j \leq q+1$ and $M \in \{j-1, s+j-1\}$, direct subtraction of the two fractions gives

$$|M/n - j/q| \leq \frac{2s+2}{n}.$$

All these ratios lie in $[0, 3/2]$. Consequently the explicit scalar discrepancy in CP28 obeys

$$V(q, s) \leq \omega_{\mathfrak{J}}((2s+2)/n) + (s/q)J_*. \quad (\text{PP35})$$

Using (PP28) at the two actual degrees in (PP34), we obtain the finite uniform estimate

$$\sup_{0 \leq j \leq q+1} \sup_{a \neq 0} \left| q^{-1} \log \frac{a^* H_0^{\text{pow}} a}{a^* H_j^{\text{pow}} a} - \mathfrak{J}(j/q) \right| \leq \delta(q, s) := 2E(q, s)/q + (n/q)\epsilon_n^G + V(q, s). \quad (\text{PP36})$$

The $j = 0$ term is zero exactly. Since $n \geq 9q/10$, the right side tends to zero whenever $s/q \rightarrow 0$. For fixed $s = \lfloor \epsilon q \rfloor$, (PP35) already gives an error tending to zero when the later limit $\epsilon \downarrow 0$ is taken. The explicit profile identity $\mathfrak{J}'(t) = \log[(1+r_t)/(1-r_t)]$, proved by the companion endpoint calculation, makes $\mathfrak{J}$ increasing and concave. Thus $\omega_{\mathfrak{J}}(h) \leq \mathfrak{J}(h)$, and its endpoint bound $\mathfrak{J}(h) = O(h \log(C/h))$ yields the stated stronger estimate

$$V(q, s) = O((s/q) \log(Cq/s) + \log(q+2)/q), \quad \limsup_{q \rightarrow \infty} \delta(q, \lfloor \epsilon q \rfloor) = O(\epsilon \log(C/\epsilon)). \quad (\text{PP37})$$

Uniform continuity alone suffices for the full convergence assertion; no microscopic derivative of a remainder is used.

PP6. Original invariant constraints and all relative spectra

Keep the original conductor space $\mathcal{C}_k$ and every original additional invariant row, denoted together by W_k . The actual kernel is $K_k = \mathcal{C}_k \cap \ker W_k$, of rank $m = 8k - 16$; it is the same coefficient space at every cutoff. The CK7 residue-error map has image dimension at most $v_0 + c_R$. On restricting that map to the already constrained kernel, its zero space G_k has actual codimension

$$c = \text{codim}_{K_k} G_k \leq \min(m, v_0 + c_R). \quad (\text{PP38})$$

This uses rank-nullity after restriction, so no condition on the rank of W_k is needed. In particular no invariant row is replaced by the conductor equation.

Fix $0 < \epsilon < 1/10$, choose the CK8 decay exponent, and then a fixed admissible contour R . Set $s = \lfloor \epsilon q \rfloor$. CK8-CK11 supplies a single injective map $A_k p = P_{\geq q-s} U_Q^{-1} p$ from G_k to the power band, valid at every cutoff. Write $\rho_k < 1$ for its uniform relative norm error, tending exponentially to zero, and C_U for the squared-norm width in the full-fibre comparison. Explicitly,

$$\|p - U_Q A_k p\|_{G_{N_j}^\sigma} \leq \rho_k \|p\|_{G_{N_j}^\sigma}, \quad e^{-C_U} \|A_k p\|_{G_{N_j}^{\text{pow}}}^2 \leq \|U_Q A_k p\|_{G_{N_j}^\sigma}^2 \leq e^{C_U} \|A_k p\|_{G_{N_j}^{\text{pow}}}^2.$$

These estimates hold before and after the complete minimum because $U_Q(y^q h) = Q_k h$ is an exact bijection between the whole fibres, with the original degree cutoff retained. The map is fixed in j ; its image need not be the whole power band or satisfy the invariant rows in those new coordinates. The vectors p on which the original metric is evaluated still lie in $G_k \subset K_k$.

Triangle inequality bounds the difference of each pair of endpoint logarithms by $2C_U + 2 \log[(1 + \rho_k)/(1 - \rho_k)]$. Finally apply the original arithmetic-Gamma form comparison, with $w_k = \log(u_k/\ell_k)$. It adds at most w_k to a logarithmic ratio. Therefore on the actual arithmetic good subspace, uniformly for all cutoffs,

$$\left| \log \frac{\|p\|_{G_{N_0}}^2}{\|p\|_{G_{N_j}}^2} - q \mathfrak{J}(j/q) \right| \leq q \eta_{k, \epsilon}, \quad (\text{PP39})$$

$$\eta_{k,\epsilon} = \delta(q, s) + \frac{2C_U + 2\log[(1 + \rho_k)/(1 - \rho_k)] + w_k}{q}, \quad \limsup_k \eta_{k,\epsilon} = O(\epsilon \log(C/\epsilon)). \quad (\text{PP40})$$

The constants retain the fixed original arithmetic data and period. This proves the finite content of CP32.

For $i \leq j$, subtract the two inequalities on the same nonzero p . Put $d_{ij} = q[\Im(j/q) - \Im(i/q)]$. Then

$$e^{d_{ij}-2q\eta} H_{G,N_j} \preceq H_{G,N_i} \preceq e^{d_{ij}+2q\eta} H_{G,N_j}. \quad (\text{PP41})$$

To relate this exact restriction to the full metric pair, choose any fixed frame inclusion $I_G : \mathbb{C}^{m-c} \rightarrow \mathbb{C}^m$. Set

$$B_{ij} = H_{K,N_j}^{-1/2} H_{K,N_i} H_{K,N_j}^{-1/2}, \quad V_j = H_{K,N_j}^{1/2} I_G (H_{G,N_j})^{-1/2}.$$

Direct multiplication proves

$$V_j^* V_j = I, \quad V_j^* B_{ij} V_j = (H_{G,N_j})^{-1/2} H_{G,N_i} (H_{G,N_j})^{-1/2}. \quad (\text{PP42})$$

Thus the entire compression has spectrum in the interval of (PP41). If the full matrix had more than c eigenvalues strictly above its upper endpoint, their spectral space would intersect the $(m - c)$ -dimensional image of V_j nontrivially, contradicting its upper Rayleigh bound. The same argument for the lower endpoint gives at most c eigenvalues below it. Hence at most $\min(m, 2c)$ full relative eigenvalues are exceptional. No simultaneous diagonalization and no orientation of the invariant rows is assumed.

Nesting and CK5-CK3 give the common coarse bound

$$0 \leq \log \lambda_a(B_{ij}) \leq 4C_1 q + w_k \leq C_0 q \quad (\text{PP43})$$

for all large k , all $i \leq j$, and every direction. The first inequality is the original source nesting. Also $0 \leq d_{ij}/q \leq \Im(1 + 1/q)$. Consequently the complete finite bound is

$$\boxed{\Omega_{ij} := \sum_{a=1}^m |\log \lambda_a(B_{ij}) - d_{ij}| \leq 2mq\eta_{k,\epsilon} + 2(v_0 + c_R)q[C_0 + \Im(1 + 1/q)]}. \quad (\text{PP44})$$

One may replace the last $2(v_0 + c_R)$ by $\min(m, 2c)$. This proves CP33 with every original invariant constraint retained.

Take $k \rightarrow \infty$ with ϵ, R fixed. Then c_R is fixed and $m = 8k - 16 \rightarrow \infty$. Divide (PP44) by mq , then let $\epsilon \downarrow 0$. This gives

$$\boxed{\sup_{0 \leq i \leq j \leq q+1} \Omega_{ij} = o(mq)}. \quad (\text{PP45})$$

There is no simultaneous growing-contour substitution or exchange of these limits. Since the logarithm of a determinant ratio is the sum of the actual relative eigenvalue logarithms, (PP45) also proves the uniform determinant profile CP7. The same argument applies to every specified cutoff-independent subspace of $\mathcal{C}_k$ whose actual rank tends to infinity: restrict the fixed residue-error map first, replace m by that rank, and keep the same finite codimension bound. Additional invariant rows can only reduce this restricted rank; they cannot increase it.

PP7. Accepted scope and precise correction

CP9, CP17, CP22-CP33, and the resulting CP7-CP8 are established by PP1-PP45 on the retained original domain. The principal added analytic detail is (PP20): a uniform barrier envelope for the moving potentials on the full positive half-line, followed in the proved order by the compactified truncation limits. The small ratio endpoint uses exact monotonicity and parity, rather than evaluating a compact-positive asymptotic at zero. The literal finite scalar remainder includes $-\frac{1}{2} \log n$, as (PP24) records. No leading coefficient changes.

Human-source formula provenance is inherited from024: R. A. Askey and R. Roy for the original Gamma equations, and T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw for the original Laguerre and Christoffel-Darboux equations. The original equation TeX, hashes, and actual reading coverage are retained in024 SOURCE_READING_USE_LEDGER.json. No new external theorem, independent native-period computation, or current sign is asserted here.

Part III

Exact elliptic profile and certified position

The original cutoff profile: exact coordinate, convergent end-point correction, and certified half-volume position

This calculation receives the cutoff function in CP3-CP8 of the supplied continuation. It preserves its original cutoff variable $t = j/q$, the physical coordinate, every original coefficient metric and every invariant row. The new variable below is a coordinate for the scalar elliptic equation only. No source metric is changed.

EP1. The original scalar and its exact coordinate map

For a modulus $0 \leq a < 1$, use the complete integrals

$$K(a) = \int_0^{\pi/2} (1 - a^2 \sin^2 \theta)^{-1/2} d\theta, \quad E(a) = \int_0^{\pi/2} (1 - a^2 \sin^2 \theta)^{1/2} d\theta. \quad (\text{EP1})$$

The original profile has $r = r_t \in (0, 1)$, $\kappa = \sqrt{1 - r^2}$, and

$$1 + t = \frac{E(\kappa)}{rK(\kappa)}, \quad J(t) = t \log \frac{1+r}{1-r} + 2 \log \frac{(1+r)K(\kappa)}{\pi}. \quad (\text{EP2})$$

Its alternative expression in CP3 follows by substituting its first equation; thus no independent constant is being fixed. Set

$$w = \frac{1-r}{1+r}, \quad \mu = w^2, \quad r = \frac{1-\sqrt{\mu}}{1+\sqrt{\mu}}, \quad \kappa = \frac{2\sqrt{w}}{1+w}, \quad k(\mu) = \frac{2K(\sqrt{\mu})}{\pi}, \quad e(\mu) = \frac{2E(\sqrt{\mu})}{\pi}. \quad (\text{EP3})$$

These formulas give mutually inverse maps between $0 < r < 1$ and $0 < \mu < 1$, preserving the positive square-root branch. The descending Landen formulas are

$$K(\kappa) = (1+w)K(w), \quad E(\kappa) = \frac{2E(w) - (1-w^2)K(w)}{1+w}. \quad (\text{EP4})$$

These classical formulas are cited to B. C. Carlson, DLMF19.8.12a-b. The original equation TeX was obtained and read in full for this calculation. We also give a proof so that the exact parameter and branch in use can be checked.

Expanding the integrand in EP1 on compact subdisks and integrating powers of sine gives

$$k(\mu) = \sum_{n \geq 0} c_n \mu^n, \quad e(\mu) = 1 - \sum_{n \geq 1} \frac{c_n \mu^n}{2n-1}, \quad c_n = \frac{\binom{2n}{n}^2}{16^n}. \quad (\text{EP5})$$

The integral of $\sin^{2n} \theta$ equals $\pi \binom{2n}{n} / 2^{2n+1}$, by integration by parts starting at $n = 0$. The binomial coefficients therefore give exactly EP5, with radius of convergence one. The second binomial series gives its stated factor $1/(2n-1)$. These are also Carlson's DLMF19.5.1-2, whose original equation TeX is retained.

Their coefficient recurrence, or differentiation of EP1 followed by integration by parts, gives

$$\frac{d}{dz} K(\sqrt{z}) = \frac{E(\sqrt{z})/(1-z) - K(\sqrt{z})}{2z}, \quad \frac{d}{dz} E(\sqrt{z}) = \frac{E(\sqrt{z}) - K(\sqrt{z})}{2z}. \quad (\text{EP6})$$

Let $z = 4w/(1+w)^2$ and $F(w) = K(\sqrt{z})/(1+w)$. Applying EP6 twice yields

$$w(1-w^2)F'' + (1-3w^2)F' - wF = 0. \quad (\text{EP7})$$

This identity follows by the quotient and chain rules; `check_profile_scalar.py` independently verifies both rational coefficients multiplying $K(\sqrt{z})$ and $E(\sqrt{z})$ vanish. For an analytic series $F = \sum a_n w^n$, EP7 gives $a_1 = 0$ and $n^2 a_n = (n-1)^2 a_{n-2}$ for $n \geq 2$. Since $F(0) = \pi/2$, its unique analytic solution is $K(w)$, by EP5. This proves the first EP4 near zero and then on the whole real interval by uniqueness for the ordinary differential equation away from zero. The identity $E(w) = (1-w^2)[wK'(w) + K(w)]$, from EP6, proves the second EP4 after substituting the first. Every branch is the real positive one fixed in EP1.

Substitution into EP2 proves the exact original-profile representation

$$t = T(\mu) = 2 \left[\frac{e(\mu)}{(1-\mu)k(\mu)} - 1 \right] = 4\mu \frac{k'(\mu)}{k(\mu)}, \quad J(t) = -\frac{t}{2} \log \mu + 2 \log k(\mu). \quad (\text{EP8})$$

The second equality for T is EP6 with its factors of two retained. In particular t is still the original cutoff fraction, not a rescaled fraction. Returning through EP3 reproduces EP2 exactly.

EP2. Unique inverse, derivatives and the positive series distribution

For each $0 < \mu < 1$, define a probability on the nonnegative integers by

$$p_n(\mu) = \frac{c_n \mu^n}{k(\mu)}. \quad (\text{EP9})$$

Every mass is positive, their sum is one, and their moments of every order converge on compact subintervals of $0 < \mu < 1$. Termwise differentiation is justified there by absolute convergence. It gives

$$T(\mu) = 4\mathbb{E}_\mu n, \quad \mu T'(\mu) = 4\text{Var}_\mu(n) > 0. \quad (\text{EP10})$$

The variance is strictly positive since both masses at zero and one are positive. This proves strict increase without subtracting nearby elliptic quantities. At zero, EP5 gives $T(\mu) = \mu + O(\mu^2)$. At one, the defining integral makes $k(\mu) \rightarrow \infty$: monotone convergence gives the divergent integral of $1/\cos \theta$. For any fixed integer L , $\sum_{n < L} p_n \rightarrow 0$, because its numerator stays bounded while $k \rightarrow \infty$. Therefore $\liminf \mathbb{E}n \geq L$, and L is arbitrary. Thus $T \rightarrow \infty$. EP8 has a unique inverse $\mu(t) \in (0, 1)$ for every $t > 0$, and this is the same r_t as EP2.

Differentiating EP8 while keeping $t = T(\mu)$, the two terms $-t/(2\mu)$ and $2k'/k$ cancel exactly. It follows that

$$f(t) := J'(t) = -\frac{1}{2} \log \mu(t) = \log \frac{1+r_t}{1-r_t} > 0, \quad J''(t) = -\frac{1}{2\mu T'(\mu)} = -\frac{1}{8\text{Var}_\mu(n)} < 0. \quad (\text{EP11})$$

The integer distribution is an auxiliary exact representation of the elliptic derivatives, with its connecting map EP8-EP10 proved above. It does not replace the arithmetic spectral measure. In particular J is increasing and strictly concave. EP8 at $\mu \downarrow 0$ gives $J(0) = 0$. For $(0 \leq x \leq y)$, the decreasing derivative implies

$$0 \leq J(y) - J(x) \leq J(y - x). \quad (\text{EP12})$$

The endpoint derivative is integrable by the expansion proved next. EP12 supplies the actual modulus of continuity used in PP35-PP37, including intervals meeting the first cutoff.

EP3. A convergent correction after the logarithmic term

EP5 gives the convergent series

$$T(\mu) = \mu + \frac{7}{8}\mu^2 + \frac{13}{16}\mu^3 + \frac{791}{1024}\mu^4 + \frac{1523}{2048}\mu^5 + O(\mu^6). \quad (\text{EP13})$$

The function is holomorphic in a neighborhood of zero and $T'(0) = 1$. The analytic inverse theorem therefore gives a convergent power series there, not only a formal asymptotic. Coefficient substitution into EP13 gives

$$\mu(t) = t - \frac{7}{8}t^2 + \frac{23}{32}t^3 - \frac{581}{1024}t^4 + \frac{1783}{4096}t^5 + O(t^6). \quad (\text{EP14})$$

For completeness, local invertibility here can also be obtained by writing $T(\mu) = \mu + \mu^2 g(\mu)$: on sufficiently small disks the equation $\mu = t - \mu^2 g(\mu)$ is a uniform contraction, whose analytic iterates converge to the inverse. Thus $\mu(t)/t$ is analytic and equals one at zero. Its logarithm with value zero at zero is analytic on a smaller disk. Substituting EP14 into the convergent logarithm series proves

$$f(t) = -\frac{1}{2} \log t + \frac{7}{16}t - \frac{43}{256}t^2 + \frac{497}{6144}t^3 - \frac{1381}{32768}t^4 + O(t^5). \quad (\text{EP15})$$

All coefficients are rational. The companion checker obtains them from finite rational polynomial operations, and checks the inverse relation through degree five. Taylor's theorem for the analytic correction gives the displayed remainder. Integrating from zero, using $J(0) = 0$, gives

$$J(t) = \frac{t}{2}(1 - \log t) + \frac{7}{32}t^2 - \frac{43}{768}t^3 + \frac{497}{24576}t^4 - \frac{1381}{163840}t^5 + O(t^6). \quad (\text{EP16})$$

The difference $J(t) - t(1 - \log t)/2$ is a convergent analytic series at zero. This strengthens the first supplied response's remainder $O(t^{5/2})$ and preserves its coefficient $7/32$. It also proves $J(h) = O(h \log(C/h))$ used to quantify the complete-jet cutoff error. No microscopic finite- k determinant coefficient is obtained by differentiating an $o(kq)$ error.

EP4. Independent outward intervals and the half-volume cutoff

The independent checker uses intervals with integer endpoints divided by 10^{90} . Each addition, subtraction, multiplication and division rounds outward with integer floor and ceiling; its square root uses integer square roots. For EP5 truncated after degree N , the positive k tail is bounded by the next term divided by $1 - \mu$, since $c_{n+1}/c_n = ((2n+1)/(2n+2))^2 < 1$. The magnitude of the e tail is bounded by this same bound divided by $2N+1$. All bisection points lie between $1/20$ and $13/20$; $N = 512$ suffices for the reported intervals. Monotonicity EP10 makes the certified signs decisive.

For $x \geq 1$, use $z = (x-1)/(x+1)$ and

$$\log x = 2 \sum_{j=0}^{L-1} \frac{z^{2j+1}}{2j+1} + R_L, \quad 0 \leq R_L \leq \frac{2z^{2L+1}}{(2L+1)(1-z^2)}. \quad (\text{EP17})$$

Integrating the geometric series proves this exact remainder. The code uses $L = 1024$ for $x = 1/\mu$ and $x = k(\mu)$, and then EP8 and EP11; no numerical value of π is needed. At $t = 1/4, 1/2, 3/4, 1$ it independently encloses $r, J, 8J, f$ strictly inside all sixteen supplied intervals. Those intervals and the narrower computed endpoints are retained in `profile_intervals_verified.json`.

There is a unique $t_{1/2} \in (0, 1)$ with $J(t_{1/2}) = J(1)/2$, by EP11. The computation first encloses the root $T(\mu) = 1$ more tightly than 10^{-55} , then bisects $J(T(\mu)) - J(1)/2$ using its outward interval, and returns through T . It certifies

$$0.28581630680480845015 < t_{1/2} < 0.28581630680480845016. \quad (\text{EP18})$$

This certifies the decimal supplied in the first response, which was not included in its interval JSON. On the original kernel, PP45 gives uniformly

$$\log \frac{\det H_{K,q-1}}{\det H_{K,q-1+[tq]}} = mJ(t) + o(kq), \quad m = 8k - 16. \quad (\text{EP19})$$

Consequently EP18 locates half of the limiting volume loss over the full original cutoff window. It is an asymptotic statement in the original fixed-period family; it is not an exact finite- k cutoff or a numerical evaluation of an assumed off-line zero. The four prescribed cutoffs still give two copies of $J(1) = C_\partial/2$, hence $mC_\partial q + o(kq)$, as in the published CK14. That earlier coefficient is preserved.

Proof provenance and receiving scope

Original incoming CP profile: `inputs/02_COMPLETE_PROOFS.md`, CP3-CP8 and CP19-CP23; both full responses are retained separately in the intake. The first response supplies its heat extension and sharper stated endpoint expansion, the second its full subquotient formulas. The analytic profile itself is verified by the independent PP proof and EP8. EP9-EP16 supplies the exact derivative representation and convergent higher correction; EP18 is a new independent certificate for the stated half-volume position.

Human reference: B. C. Carlson, *Elliptic Integrals*, DLMF Chapter19, original equation TeX19.8.12a,19.8.12b,19.5.1,19.5.2. These four formula sources were read, downloaded intact and hashed. No whole-book reading is claimed. The full elementary derivations used here are EP1-EP17. The original arithmetic and Gamma objects are connected by the retained024 CK/PJ/EL proofs and the new PP proof, all included in the cumulative source bank at publication.

Part IV

Original measured phase and complete heat

Arbitrary-cutoff phase and measured-heat profile in the original metrics

This proof checks CP37–51 and completes the measured-heat argument appearing only in the first part of the supplied continuation. The original physical scale, complete observation, full kernel frame, zero-word factors and cutoff indexing are retained. The stronger heat estimates below follow from an exact convolution, and do not assign a sign to the separate arithmetic-current insertion.

The scalar power-jet profile CP9/27/33 and the evaluated function $\mathfrak{J}$ are established in the companion independent derivations. Their actual finite errors enter here explicitly; they are not assumed to vanish without those calculations.

1. Original data and the image polynomial map

Retain

$$q = (k+1)^2, \quad q' = (k-7)^2, \quad \Delta = q - q' = 16k - 48, \quad m = 8k - 16, \quad k \equiv 1 \pmod{4},$$

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta] = D_k(y)O_k(y), \quad E_k = \mathbb{C}[y]/(Q_k), \quad M[p] = [yp], \quad T = D_k(M). \quad (\text{PH1})$$

The physical coordinate is $s = k/2 + iy$, with the original fixed period, branch, arithmetic unit and $0 < \delta < 1/2, \gamma > 2$. The degrees are $\deg D_k = q'$, $\deg O_k = \Delta$. The original complete-source quotient metric at degree N is $G_N > 0$; no new source is substituted. For the original onto observation Λ , retain its complete fixed kernel inclusion I_K , $K = \ker \Lambda$, and

$$H_{K,N} = I_K^* G_N I_K, \quad Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_{B,N}. \quad (\text{PH2})$$

Use $N_j = q - 1 + j$, $0 \leq j \leq q+1$, $t_j = j/q$. In particular $j = q+1$ is the actual cutoff $2q$. All subsequent heat averages explicitly use only $0 \leq j < q$, as in the supplied paste; this does not remove the extra original cutoff from the determinant or phase statements.

Write $\|p\|_{c,q} = \sum_a q^a |[y^a]p|$, and $\rho = k\sqrt{\delta^2 + \gamma^2}/q$. The finite coefficient estimates below have the stated guard $0 < \rho < 1$, which holds for all sufficiently large k at the retained fixed data. The exact polynomial-part map and its inverse are

$$U_Q f = [Q_k(y)y^{-q}f(y)]_+, \quad U_Q^{-1}p = [y^q p(y)/Q_k(y)]_+.$$

They are unit triangular on degree $< q$. More generally, multiplication and polynomial-part division at infinity give

$$U_Q(f + y^q h) = U_Q f + Q_k h, \quad U_Q^{-1}(D_k a) = \left[\frac{y^q a(y)}{O_k(y)} \right]_+. \quad (\text{PH3})$$

The equalities retain every relation column and every affine fibre.

For completeness, put $\widehat{D}(x) = q^{-q'} D_k(qx)$, $\widehat{O}(x) = q^{-\Delta} O_k(qx)$, $\widehat{p}(x) = p(qx)$, $\widehat{a}(x) = q^{q'} a(qx)$. Every root of the two monic polynomials has modulus at most ρ , and $\widehat{p} = \widehat{D} \widehat{a}$. Reversing coefficients, the inverse leading triangular block of multiplication by $\widehat{D}$ consists of the complete homogeneous coefficients of its roots. Their total absolute sum is at most $(1 - \rho)^{-q'}$; hence

$$\|\widehat{a}\|_1 \leq (1 - \rho)^{-q'} \|\widehat{p}\|_1. \quad (\text{PH4})$$

Indeed the coefficient of order h in the reciprocal reversed polynomial is at most $(q'^{h-1})\rho^h$, and summing the full nonnegative series gives $(1 - \rho)^{-q'}$.

For any integer $s > \Delta$, the reciprocal coefficients of $\widehat{O}$ are bounded by $(\Delta^{h-1})\rho^h$. A term of $[x^q \widehat{a}(x)/\widehat{O}(x)]_+$ below degree $q - s$ has depth at least $s - \Delta$, and every depth contributing to its

polynomial part is at most $q - 1$. Summing all these terms, using $\sum_{h=0}^{q-1} \binom{\Delta+h-1}{h} = \binom{\Delta+q-1}{q-1} \leq 2^{q+\Delta}$, proves the full inverse-tail bound

$$\|P_{< q-s} U_Q^{-1} p\|_{c,q} \leq \epsilon_{\text{tail}}(q, s) \|p\|_{c,q}, \quad \epsilon_{\text{tail}} = (1 - \rho)^{-q'} 2^{q+\Delta} \rho^{s-\Delta}, \quad p \in W := D_k \mathcal{P}_{< \Delta}. \quad (\text{PH5})$$

This proves the more direct full-image estimate in the paste. Similarly, the elementary symmetric coefficients of $\hat{D}$ give

$$\|P_{< q-s/2} p\|_{c,q} \leq (1 - \rho)^{-q'} 2^{q'} \rho^{s/2-\Delta} \|p\|_{c,q} \quad (\text{PH6})$$

whenever $s/2 > \Delta$; the looser bound also holds when its displayed right-hand multiplier is at least the full multiplication estimate. Integer depths only improve the real exponent written in PH6. Thus CP38 has a valid finite root-product proof, with its degree and radius guards explicit.

Here is a complete finite norm-transfer guard. In the coefficient coordinates $x_a = q^a [y^a]p$, let $\tilde{G}_N$ represent G_N , and set

$$a_* = \frac{1}{q} \min_{q-1 \leq N \leq 2q} \lambda_{\min}(\tilde{G}_N), \quad b_* = \max_{q-1 \leq N \leq 2q} \lambda_{\max}(\tilde{G}_N), \quad M_Q = (1 + \rho)^q.$$

Then $a_* \|p\|_{c,q}^2 \leq \|p\|_{G_N}^2 \leq b_* \|p\|_{c,q}^2$, since $\|x\|_2 \leq \|x\|_1 \leq \sqrt{q} \|x\|_2$. Elementary symmetric coefficients of Q_k give $\|U_Q\|_{c,q} \leq M_Q$. Define

$$\vartheta = \sqrt{b_*/a_*} M_Q \epsilon_{\text{tail}}. \quad (\text{PH7})$$

On the explicit guard $\vartheta < 1$, the fixed map $A_W p = P_{\geq q-s} U_Q^{-1} p$ is injective on all of W , and

$$(1 - \vartheta) \|p\|_{G_N} \leq \|U_Q A_W p\|_{G_N} \leq (1 + \vartheta) \|p\|_{G_N} \quad \text{for every } N. \quad (\text{PH8})$$

The proof is the triangle inequality applied to $p - U_Q A_W p = U_Q P_{< q-s} U_Q^{-1} p$, followed by PH5 and the two coefficient bounds. If $A_W p = 0$, PH8 forces $p = 0$. This is a map on every original image direction; no exceptional residue subspace is used.

For the following complete source comparison impose the further finite band guard $\Delta < s \leq q/10$. The asymptotic choice below uses $0 < \epsilon < 1/10$.

The complete high-power source comparison CK3 and CK10-11 supplies the same actual two constants at every cutoff and every band vector g :

$$a_k^{\text{src}} \|g\|_{\text{pow},N}^2 \leq \|U_Q g\|_{G_N}^2 \leq b_k^{\text{src}} \|g\|_{\text{pow},N}^2, \\ a_k^{\text{src}} = \ell_k e^{-C}, \quad b_k^{\text{src}} = u_k e^C, \quad \chi_k = \log(b_k^{\text{src}}/a_k^{\text{src}}) = 2C + \log(u_k/\ell_k).$$

Its terms are the complete two-region U_Q comparison and the actual arithmetic/Gamma form comparison; no observation row is substituted for a form. Let $q\delta(q, s)$ be the independently established all-vector error in CP27-28:

$$\left| \log \frac{\|g\|_{\text{pow},N_0}^2}{\|g\|_{\text{pow},N_j}^2} - q\mathfrak{J}(t_j) \right| \leq q\delta(q, s).$$

Combining these inequalities with PH8, keeping both endpoints, gives on all W

$$\left| \log \frac{\|p\|_{G_{N_0}}^2}{\|p\|_{G_{N_j}}^2} - q\mathfrak{J}(t_j) \right| \leq q\zeta(q, s), \quad \zeta(q, s) = \delta(q, s) + \frac{\chi_k + 2 \log((1 + \vartheta)/(1 - \vartheta))}{q}. \quad (\text{PH9})$$

To verify the last constant directly, at each cutoff the logarithm $\log(\|p\|_{G_N}^2 / \|A_W p\|_{\text{pow},N}^2)$ belongs to $[\log a_k^{\text{src}} - 2 \log(1 + \vartheta), \log b_k^{\text{src}} - 2 \log(1 - \vartheta)]$. Its possible two-cutoff difference is exactly bounded by the width used in PH9.

For fixed $s = \lfloor \epsilon q \rfloor$, PH5 has logarithm $-c\epsilon q \log k + O(q)$, whereas the complete original coefficient comparison has logarithm $O(q)$. Thus $\vartheta \rightarrow 0$. The proved power estimate gives $\limsup \delta(q, s) = O(\epsilon \log(C/\epsilon))$; the complete source comparison has $\chi_k = o(q)$. Taking $k \rightarrow \infty$ first and then $\epsilon \downarrow 0$, or taking the exact infimum of PH9 over its finitely many admitted integer s , proves a common $\zeta_k \rightarrow 0$. This is CP39 with the full norm error displayed. It uses the companion power and source estimates at their proved status, not a new equilibrium assumption.

2. Full positive-word centre and paired compressed energies

Let $\pi_{\partial} : E_k \rightarrow \mathbb{C}^{\Delta}$ be the original boundary-value map, with kernel $V = \ker T$, and $B_N^{\partial} = (\pi_{\partial} G_N^{-1} \pi_{\partial}^*)^{-1}$. Let $I_W : \mathbb{C}^{\Delta} \rightarrow W$ be the fixed boundary-value frame, so that $G_{W,N} = I_W^* G_N I_W$ and $T = I_W D_{\partial} \pi_{\partial}$, where D_{∂} contains the unchanged values $D_k(\omega)$. The minimum section $S_N = G_N^{-1} \pi_{\partial}^* B_N^{\partial}$ satisfies

$\pi_\partial S_N = I$ and $S_N^* G_N S_N = B_N^\partial$; its image is $V^{\perp_{G_N}}$. Therefore the positive energies of $H_N = G_N^{-1} T^* G_N T$ are exactly the generalized eigenvalues of

$$F_N^W = D_\partial^* G_{W,N} D_\partial \quad \text{against} \quad B_N^\partial. \quad (\text{PH10})$$

This proves CP37 with its actual domain, codomain and metric. The other q' full-word energies are exactly zero.

Let $c_0 = 2q \log q + [2 \log(4/\pi) - 2]q$, and let $E_0 = \max_{a \leq \Delta} |\log \lambda_{a,N_0}^+ - c_0|$ be the original initial-word error. The complete boundary comparison gives $e^{-\omega_k} B_{N_0}^\partial \preceq B_{N_j}^\partial \preceq B_{N_0}^\partial$, where $\omega_k = \log(b_\partial/a_\partial)$, and the second inequality is source nesting. By PH9 and generalized min-max,

$$-E_0 - q\zeta_k \leq \log \lambda_{a,N_j}^+ - c_k(t_j) \leq E_0 + q\zeta_k + \omega_k, \quad c_k(t) = c_0 - q\mathfrak{J}(t). \quad (\text{PH11})$$

One may replace ω_k at j by the actual $\log \lambda_{\max}((B_{N_j}^\partial)^{-1/2} B_{N_0}^\partial (B_{N_j}^\partial)^{-1/2})$. Thus the single finite error $e_k = E_0/q + \zeta_k + \omega_k/q$ proves CP40 at all $q+2$ actual cutoffs. The initial error obeys $E_0 \leq \mathcal{E}_k = O_{h,\varpi}(k \log(q+2))$ by 019 FW8-9, including their explicit constants. For all $q-1 \leq N \leq 2q$, put $R_k = k\sqrt{\delta^2 + \gamma^2}$ and retain

$$A_{2q}^\partial = \frac{\Delta e^2}{\sqrt{2\pi}} (2q+1)^{3/2} (4q+1)^{R_k+1}, \quad B_k^\partial = \Delta \sqrt{2\pi} \left(\frac{2\Delta + R_k}{2\delta} \right)^{2(\Delta-1)}.$$

The complete boundary evaluation and Lagrange-lift proofs 018 IC12-15 give the native constants $a_\partial = \ell_k/A_{2q}^\partial$, $b_\partial = u_k B_k^\partial$. Consequently $\omega_k = \log[(u_k/\ell_k) A_{2q}^\partial B_k^\partial] = O_{h,\varpi}(k \log(q+2)) = o(q)$. This uses the actual source ratio in CK3, whose logarithm is $O(k + \log q)$, and the displayed boundary factors. It proves both required vanishing errors on the entire cutoff window; hence $e_k \rightarrow 0$. In particular $\eta(t) = c_k(t)/q - 2 \log q = 2 \log(4/\pi) - 2 - \mathfrak{J}(t)$, including the original endpoints.

For the kernel put $F_{K,N} = (T I_K)^* G_N (T I_K)$. On the native transverse family it is positive and its compressed energies are the eigenvalues $\alpha_{a,N}$ of $A_N = H_{K,N}^{-1/2} F_{K,N} H_{K,N}^{-1/2}$. For arbitrary positive P, Q, F , define $R = P^{1/2} \max(I, P^{-1/2} Q P^{-1/2}) P^{1/2}$. Then $R \succeq P, Q$. Generalized min-max puts the ordered log spectrum of (F, R) below those of (F, P) and (F, Q) . Pair each of the latter lists through the lower list, use the triangle inequality, and sum the exact determinant differences to get

$$\sum_a |\log \lambda_a(F, P) - \log \lambda_a(F, Q)| \leq \log \frac{\det R}{\det P} + \log \frac{\det R}{\det Q} = \sum_a |\log \lambda_a(P^{-1/2} Q P^{-1/2})|. \quad (\text{PH12})$$

This proves CP42 without common eigenvectors.

Restrict PH9 to the fixed image $T I_K$, cancel the common scalar $\exp[-q(\mathfrak{J}(t_j) - \mathfrak{J}(t_i))]$ between the two forms, and apply PH12. With the complete relative-kernel discrepancy $\Omega_{i,j}$ of CP8, one obtains

$$D_{i,j} := \sum_{a=1}^m |\log \alpha_{a,N_i} - \log \alpha_{a,N_j}| \leq \Omega_{i,j} + 2mq\zeta_k. \quad (\text{PH13})$$

This proves CP43; its $o(mq)$ conclusion uses the independently proved CP8 and PH9. If $Z = K \cap V \neq 0$, replace $H_{K,N}$ by its actual attained quotient on K/Z , retain $p = \text{rank}(T|_K)$ positive energies and $m-p$ zero factors, and use the same fixed image form. In a fixed split frame $[I_Z, I_1] = I_K S$, this quotient is $H_{11,N} - H_{01,N}^* H_{00,N}^{-1} H_{01,N}$. Both the full $H_{00,N}$ and the fixed frame volume remain in the full kernel determinant. PH12 applies to this positive quotient; its relative log deviation is controlled by the original relative-spectrum compression theorem. No inverse or logarithm of a zero energy is taken.

3. Finite determinant, phase and arbitrary-cutoff errors

For each cutoff define the original measured resolvent

$$\mathcal{Y}_N(z) = \Lambda z(zI + H_N)^{-1} L_N.$$

The actual isometries $J_{B,N} = L_N Q_{B,N}^{-1/2}$ and $J_{K,N} = I_K H_{K,N}^{-1/2}$ are orthogonal complements. Full block elimination therefore gives

$$\det \mathcal{Y}_N(z) = \frac{\det(I_m + A_N/z)}{\det(I_q + H_N/z)}. \quad (\text{PH14})$$

It is valid also when A_N has zeros. Both dimensions and all zero factors remain in the ratio. Analytic logarithms are continued from infinity on the slit plane.

For a sharper finite error than a uniform maximum, set

$$E_j^W = \sum_{a=1}^{\Delta} |\log \lambda_{a,N_j}^+ - c_k(t_j)|, \quad \mathcal{B}_{i,j} = D_{i,j} + E_i^W + E_j^W \leq D_{i,j} + 2\Delta q e_k. \quad (\text{PH15})$$

The derivative of $a \mapsto \log(1 + e^a/z)$ is between zero and one for $z > 0$. Apply its Lipschitz bound to every matched positive energy in PH14:

$$\sup_{z>0} \left| \log \frac{\det \mathcal{Y}_{N_i}(z)}{\det \mathcal{Y}_{N_j}(z)} + \Delta \log \frac{z + e^{c_k(t_i)}}{z + e^{c_k(t_j)}} \right| \leq \mathcal{B}_{i,j}. \quad (\text{PH16})$$

For $|\arg z| \leq \pi - \delta$, $0 < \delta \leq \pi$, the same expression in the continued analytic logarithms is bounded by $\csc(\delta/2)\mathcal{B}_{i,j}$. Indeed $|z + s| \geq \sin(\delta/2)(|z| + s)$ implies $|s/(z + s)| \leq \csc(\delta/2)$. This proves CP45 and its complex extension.

At imaginary frequency, every scalar full resolvent has positive real part and nonnegative imaginary part. Compression preserves the strict positive real part, so the principal matrix logarithm of $Q_{B,N}^{1/2} \mathcal{Y}_N(i\omega) Q_{B,N}^{-1/2}$ exists. Its trace gives the unwrapped phase anchored at zero as $\omega \rightarrow \infty$. The determinant identity then proves

$$\Theta_N(\omega) = \sum_{a=1}^{\Delta} \arctan(\lambda_{a,N}/\omega) - \sum_{a=1}^p \arctan(\alpha_{a,N}/\omega). \quad (\text{PH17})$$

This is CP46, with the zero-intersection extension explicit.

Let $f(x) = \arctan(e^x)$; $f'(x) = 1/(2 \cosh x) > 0$ and $\int_{\mathbb{R}} f'(x) dx = \pi/2$. For any real a, b , Tonelli applied to $|f(a-x) - f(b-x)| = \int_{\min(a,b)}^{\max(a,b)} f'(t-x) dt$ proves its exact integral $(\pi/2)|a-b|$. Therefore

$$\int_0^\infty |\Theta_{N_i} - \Theta_{N_j} - \Delta[\arctan(e^{c_i}/\omega) - \arctan(e^{c_j}/\omega)]| \frac{d\omega}{\omega} \leq \frac{\pi}{2} \mathcal{B}_{i,j}. \quad (\text{PH18})$$

The signed integral is exactly

$$\frac{2}{\pi} \int_0^\infty (\Theta_{N_i} - \Theta_{N_j}) \frac{d\omega}{\omega} = \log \frac{\det_+ H_{N_i}}{\det_+ H_{N_j}} - \log \frac{\det_+ A_{N_i}}{\det_+ A_{N_j}}, \quad (\text{PH19})$$

and hence differs from $\Delta q[\mathfrak{J}(t_j) - \mathfrak{J}(t_i)]$ by at most $\mathcal{B}_{i,j}$. The equal ranks at both cutoffs make the two frequency ends integrable. The finite equality precedes any asymptotic. The original four signs add the two pairs $(0, q)$ and $(1, q+1)$, preserving both adjacent degrees.

At $\omega = q^{2q}e^{bq}$, the exact scalar step error is

$$\int_{\mathbb{R}} |\arctan(e^{q(a-b)}) - (\pi/2)\mathbf{1}_{b < a}| db = \frac{2G_{\text{Cat}}}{q}, \quad G_{\text{Cat}} = \int_0^1 \frac{\arctan x}{x} dx.$$

Split the integral at $b = a$, use $\arctan x + \arctan(1/x) = \pi/2$, and substitute $x = e^{-q|a-b|}$ to prove the constant. Thus for $i \leq j$

$$\int_{\mathbb{R}} \left| \frac{\Theta_{N_i}(q^{2q}e^{bq}) - \Theta_{N_j}(q^{2q}e^{bq})}{k} - \frac{\pi\Delta}{2k} \mathbf{1}_{\eta(t_j) < b < \eta(t_i)} \right| db \leq \frac{\pi\mathcal{B}_{i,j}}{2kq} + \frac{4\Delta G_{\text{Cat}}}{kq}. \quad (\text{PH20})$$

This proves CP50 with its exact constants. The common error is $o(1)$, so for fixed $s < t$ the limiting height is 8π .

For the step interpolation $j(t) = \lfloor qt \rfloor$, $0 \leq t < 1$, the extra two-variable comparison to $\eta(t)$ has integral at most

$$\frac{\pi\Delta}{2k} \int_0^1 [\mathfrak{J}(t) - \mathfrak{J}(\lfloor qt \rfloor/q)] dt \leq \frac{\pi\Delta \mathfrak{J}(1)}{2kq}. \quad (\text{PH21})$$

The bound follows by monotonicity and upper/lower Riemann sums. Integrate PH20 in t , then use PH21. Distributional differentiation against any smooth compactly supported test in $0 < t < 1$ gives the limit $8\pi f(t)\delta_{b=\eta(t)}$, because $\eta'(t) = -f(t)$. This proves CP51 without differentiating a pointwise error.

4. Complete measured heat and its exact angle defect

For physical heat time $\tau > 0$, retain

$$\mathcal{Z}_N(\tau) = \Lambda e^{-\tau H_N} L_N, \quad \Gamma_N = J_{K,N}^* G_N P_{V^\perp G_N} J_{K,N}, \quad \vartheta_N^\Gamma = \text{Tr} \Gamma_N. \quad (\text{PH22})$$

The observed operator is similar by its actual $Q_{B,N}^{1/2}$ to $J_{B,N}^* G_N e^{-\tau H_N} J_{B,N}$. The complete orthogonal resolution therefore gives

$$\text{Tr}_B \mathcal{Z}_N(\tau) = (q - m - \Delta) + \sum_{\lambda > 0} e^{-\tau \lambda} + d_N(\tau), \quad d_N(\tau) = \text{Tr}[J_{K,N}^* G_N (I - e^{-\tau H_N}) J_{K,N}]. \quad (\text{PH23})$$

Indeed the full trace is $q - \Delta + \sum_{\lambda > 0} e^{-\tau \lambda}$, and the kernel compression trace is $m - d_N(\tau)$. This proves the paste's identity including every zero factor, even at a nonzero intersection $K \cap V$.

For actual positive spectral bounds l_N, u_N , functional calculus gives

$$(1 - e^{-\tau l_N}) \Gamma_N \preceq J_{K,N}^* G_N (I - e^{-\tau H_N}) J_{K,N} \preceq (1 - e^{-\tau u_N}) \Gamma_N. \quad (\text{PH24})$$

Taking traces retains the same constants. The bounds are attained at $H_N = l_N P$ or $u_N P$. Also $d'_N(\tau) = \text{Tr}(J_K^* G_N H_N e^{-\tau H_N} J_K) \geq 0$. Thus its range is $[0, \vartheta_N^\Gamma]$, and the exact observed zero-energy trace is

$$\text{Tr}_B(\Lambda P_V^{G_N} L_N) = q - m - \Delta + \vartheta_N^\Gamma. \quad (\text{PH25})$$

The subtracted integer in the supplied observable is not declared equal to this generally noninteger mass.

Use precisely the declared physical scale and exponent:

$$\tau_k(\xi) = q^{-2q} e^{-q\xi}, \quad u_{a,j} = q^{-1} \log \lambda_{a,N_j} - 2 \log q, \quad 0 \leq j < q. \quad (\text{PH26})$$

Then $\tau_k(\xi) \lambda_{a,N_j} = e^{q(u_{a,j} - \xi)}$. Keep the original integer-subtracted observable as the primary one:

$$F_k(\xi) = \frac{1}{q\Delta} \sum_{j=0}^{q-1} [\text{Tr}_E e^{-\tau_k(\xi) H_{N_j}} - q'],$$

$$M_k(\xi) = \frac{1}{q\Delta} \sum_{j=0}^{q-1} [\text{Tr}_B \mathcal{Z}_{N_j}(\tau_k(\xi)) - (q - m - \Delta)]. \quad (\text{PH27})$$

Its complete connecting map is

$$M_k = F_k + D_k^{\text{ang}}, \quad D_k^{\text{ang}}(\xi) = \frac{1}{q\Delta} \sum_j d_{N_j}(\tau_k(\xi)), \quad 0 \leq D_k^{\text{ang}}(\xi) \leq \alpha_k := \frac{1}{q\Delta} \sum_j \vartheta_{N_j}^\Gamma. \quad (\text{PH28})$$

The defect decreases in ξ , from α_k to zero. Hence $M_k(-\infty) = \alpha_k$ and $M_k(+\infty) = 1$. The sharper frequency-dependent bounds are those of PH24, summed at the actual cutoffs.

The inherited full-angle bound, retained as a finite sum, is

$$\gamma_{v+r,N} \leq \min(1, \mathfrak{B}_k r^{-2q'}) \implies \vartheta_N^\Gamma \leq \min \left(m, v + \sum_{r=1}^{(m-v)_+} \min(1, \mathfrak{B}_k r^{-2q'}) \right) =: T_k^{\Gamma, \text{fin}}. \quad (\text{PH29})$$

For $v \geq m$, use $T_k^{\Gamma, \text{fin}} = m$. For $q' > 1/2$, putting $J = \lceil \mathfrak{B}_k^{1/(2q')} \rceil$ and integrating the decreasing tail gives the further bound $\min(m, v + J[1 + (2q' - 1)^{-1}])$. Thus $\alpha_k \leq T_k^{\Gamma, \text{fin}}/\Delta$. Here is the exact provider and its map to this primary angle operator. The complete source is the published 024 original measured-word proof, equations (10)–(24); its full conductor-to-invariant compression and trace sum are at equations (21)–(24). The following reconstruction retains its constants and proves why it applies at every cutoff here.

In full root-value coordinates let $\mathcal{J}_\partial b = (Fb, b)$, where $F = -A_\mathcal{T}^{-1} A_\partial$ is the actual conductor graph. Its image is V_A , the complete conductor kernel, which contains K . Put

$$H_{A,N} = \mathcal{J}_\partial^* G_N^{\text{val}} \mathcal{J}_\partial, \quad B_K = \pi_\partial I_K^{\text{val}}, \quad I_K^{\text{val}} = \mathcal{J}_\partial B_K. \quad (\text{PH29a})$$

The full Vandermonde V gives $G_N^{\text{val}} = V^{-*} G_N V^{-1}$ and $I_K^{\text{val}} = V I_K$, so these equalities preserve the actual metric. Let $v = \text{ord}_0 E_A$, where the unchanged conductor symbol is $E_A(w) = \sum_\beta a_\beta e^{b_\beta w}$. The polynomials of degree below v give a v -dimensional subspace $\mathcal{L} \subset V_A$. Choose a coisometry

$\Pi : \mathbb{C}^\Delta \rightarrow \mathbb{C}^{\Delta-v}$ with $\ker \Pi = \pi_\partial \mathcal{L}$. The covariance of its full attained quotient is exactly $\Pi H_{A,N}^{-1} \Pi^*$, by minimizing a positive quadratic form on each affine fibre.

For a row z of this full coisometry define the original functional

$$F_z(w) = \sum_{\beta \in \partial} z_\beta e^{\beta w}, \quad A_z(w) = e^{-(k/2-4)w} F_z(w)/E_A(w), \quad \ell_z[f] = [f(-i\partial_w)A_z(w)]_{w=0}.$$

The first v numerator derivatives vanish, so the displayed quotient is holomorphic at zero. The exact conductor identity is $\ell_z[\mathcal{T}_A p] = z \mathcal{V} p$. On every source representative with value in V_A , $\mathcal{T}_A p$ belongs to the entire lower-root ideal $I_N = Q_- \mathcal{P}_{N-v-q'}$. The complete weighted conductor norm in the retained proof is

$$\mathfrak{M}_k = \max\{1, e^{1+2\pi\gamma} A_{\text{abs}}(8q+1)^{1/4}(2q+1)(4q+1)^{4\delta}\}, \quad A_{\text{abs}} = \sum_{\beta} |a_\beta|,$$

and it proves $\|\mathcal{T}_A p\|_\sigma \leq \mathfrak{M}_k \ell_k^{-1/2} \|p\|_{\mu_k}$. Let C_N^{ideal} be the complete covariance of the displayed functionals on I_N with its full Gamma norm. Taking the supremum of the squared functional over every identical source fibre, then the minimum source norm, proves

$$\Pi H_{A,N}^{-1} \Pi^* \preceq (\mathfrak{M}_k^2 / \ell_k) C_N^{\text{ideal}}. \quad (\text{PH29b})$$

In detail, for a row combination z , the numerator $|z \mathcal{V} p|^2$ equals $|\ell_z[\mathcal{T}_A p]|^2$, which is at most its ideal dual norm squared times $\mathfrak{M}_k^2 \|p\|_{\mu_k}^2 / \ell_k$. Supremizing over all representatives proves the quadratic-form inequality. This retains the full conductor quotient; it is not a replacement of K .

Here are finite constants for all ideal eigenvalues, not just asymptotic notation. Set $R_h = \sqrt{\delta^2 + \gamma^2}$, $B_A = \max |b_\beta|$, $d_{\text{cond}} = |E_A^{(v)}(0)|/v! > 0$, $\bar{B}_A = \max(1, B_A)$, and

$$\begin{aligned} a_A &= \frac{\log^+(A_{\text{abs}}/d_{\text{cond}})}{\log 2}, \quad b_A = \frac{4B_A}{\log 2}, \quad L_A = b_A + 9, \quad j_A = \lceil \max\{2(a_A + 1), a_A + 1 + 4L_A\} \rceil, \\ \rho_A &= \min \left\{ 1, \frac{(v+1)! d_{\text{cond}}}{2A_{\text{abs}} \bar{B}_A^{v+1} e^{\bar{B}_A}} \right\}, \quad C_{\text{an}} = \frac{\max(A_{\text{abs}}, d_{\text{cond}})^2}{d_{\text{cond}}^3}, \quad c_\sigma = \Gamma(1/4)^2/(2\pi), \quad M_{\text{max}} = 2q - v - q', \\ \mathfrak{A}_k &= (M_{\text{max}} + 1) 10^{M_{\text{max}}} 2^{q'} e^{\pi q/2} \sqrt{\frac{1+16q^2}{c_\sigma q}}, \quad \mathfrak{F}_k = \mathfrak{A}_k C_{\text{an}} \sqrt{\Delta} e^{2R_h q + (8+4B_A)k} (R_h + 2)^{q'}, \\ M_{0,k} &= 2\sqrt{\Delta} \rho_A^{-v} d_{\text{cond}}^{-1} e^{(4+kR_h)\rho_A}, \quad \mathfrak{F}_{0,k} = \mathfrak{A}_k M_{0,k} (kR_h/q + 2/\rho_A)^{q'} (2/\rho_A)^{M_{\text{max}}}, \\ \mathfrak{U}_k &= \max\{\mathfrak{F}_k^2 (4L_A)^{2q'}, \mathfrak{F}_{0,k}^2 j_A^{2q'}\}. \end{aligned} \quad (\text{PH29c})$$

Their finite domain includes $v < \Delta$, $N - v \geq q'$, and $q \geq 2kR_h$; eventually all hold in the fixed-data family. The full-strip proof in equations (14)–(20) of the cited source is as follows. Jensen's formula bounds the number of conductor zeros in $|w| < 2R$, with multiplicity, by $a_A + b_A R$. Imposing their numerator cancellations costs at most that codimension. The radius- $2R$ Blaschke quotient, followed by the zero-free Harnack bound on radius R , gives $\sup_{|w| \leq R} |A_z(w)| \leq C_{\text{an}} \sqrt{\Delta} e^{(2kR_h + 8 + 4B_A)R} \|z\|$. The Gamma product lower bound $\sigma(y) \geq c_\sigma e^{-\pi|y|/2}/(1+4y^2)$, all q' root factors on $[q, 2q]$, and the full shifted Legendre basis give $\|\text{coef } p(q)\|_1 \leq \mathfrak{A}_k q^{-q'} \|Q - p\|_\sigma$. Apply Cauchy's derivative estimate through every degree $N - v \leq 2q$. For $2 \leq R \leq k$, the resulting dual bound is $\|\ell_z|_{I_N}\| \leq \mathfrak{F}_k R^{-q'} \|z\|$. The Taylor remainder of $E_A(w)/w^v$ on ρ_A is at most $d_{\text{cond}}/2$, so unrestricted rows have dual norm at most $\mathfrak{F}_{0,k}$. For $j \geq j_A$, take $R_j = (j - 1 - a_A)/(2L_A)$. Then $R_j \geq 2$, $R_j \geq j/(4L_A)$, and $a_A + b_A R_j < j$. The full strip has dimension $g = \Delta - v < 16k$, hence $R_j < g/(2L_A) < k$; this checks the extended rank range explicitly. Min-max on the cancellation subspace and the unrestricted estimate at the first indices now prove

$$\lambda_j^\downarrow(C_N^{\text{ideal}}) \leq \mathfrak{U}_k j^{-2q'}, \quad 1 \leq j \leq \Delta - v, \quad \log \mathfrak{U}_k = O_{h,\varpi}(q). \quad (\text{PH29d})$$

The full source proofs of the Gamma, Legendre, disk and derivative estimates are retained in 019 AS12-25 and the cited 024 extension. Only the strip dimension changes from the earlier invariant-row use; PH29c-d include that change in both square-root factors and the upper radius guard.

Now form the whole-conductor angle and the exact invariant-kernel isometry

$$\Gamma_{A,N} = H_{A,N}^{-1/2} B_N^\partial H_{A,N}^{-1/2}, \quad U_{K,N} = H_{A,N}^{1/2} B_K H_{K,N}^{-1/2}.$$

PH29a gives $U_{K,N}^* U_{K,N} = I_m$, and direct multiplication gives

$$\Gamma_N = U_{K,N}^* \Gamma_{A,N} U_{K,N}. \quad (\text{PH29e})$$

The boundary minimum is no larger than the norm of its particular conductor lift, so $0 \preceq \Gamma_{A,N} \preceq I_\Delta$. The full boundary interpolation above gives $B_N^\partial \preceq u_k B_k^\partial I$. Thus min-max, followed by the coisometry interlacing that deletes exactly v directions, and then PH29b-d, yields

$$\gamma_{v+j,N}^\downarrow \leq u_k B_k^\partial \lambda_{v+j}^\downarrow (H_{A,N}^{-1}) \leq u_k B_k^\partial \lambda_j^\downarrow (\Pi H_{A,N}^{-1} \Pi^*) \leq \mathfrak{B}_k j^{-2q'}, \quad \mathfrak{B}_k = \max\{1, (u_k/\ell_k) B_k^\partial \mathfrak{M}_k^2 u_k\}. \quad (\text{PH29f})$$

There is no conductor-graph or invariant-frame condition-number factor. All constants are uniform for $q-1 \leq N \leq 2q$. Their displayed formulas give $\log \mathfrak{B}_k = O(q)$; the $B_k^\partial, \mathfrak{M}_k$ and u_k/ℓ_k terms contribute $o(q)$. Since $q'/q \rightarrow 1$ and v is fixed, PH29's J is bounded and $\sup_N \text{Tr} \Gamma_N = O_{h,\varpi}(1)$. This proves $\alpha_k = O(1/\Delta) \rightarrow 0$ in the original metric. The finite identities PH22-28 themselves do not require this eventual estimate.

5. The exact Gumbel convolution and an improved finite error

Let E_{exp} have scalar probability density e^{-x} , $x > 0$, and set $G_{\text{Gu}} = -\log E_{\text{exp}}$. Then

$$\mathbb{P}(G_{\text{Gu}} \leq x) = e^{-e^{-x}}, \quad e^{-e^{q(u-\xi)}} = \mathbb{P}(u + G_{\text{Gu}}/q \leq \xi). \quad (\text{PH30})$$

This auxiliary scalar probability represents the physical heat function exactly; it does not replace an arithmetic source. Consequently F_k is the distribution function of $(q\Delta)^{-1} \sum_{j,a} \delta_{u_{a,j}}$ convolved with the law of G_{Gu}/q . The q' exact zero modes were subtracted before forming this probability.

Put $\delta_k = \max_{j < q,a} |u_{a,j} - \eta(t_j)| \leq e_k$, $f_* = f(1) > 0$, and $L_* = 1/f_*$. The target is the pushforward of Lebesgue probability on $[0, 1]$ by the actual evaluated decreasing function $\eta(t) = \eta(0) - \mathfrak{J}(t)$:

$$\mathcal{H}(\xi) = \text{Leb}\{t : \eta(t) \leq \xi\} = \begin{cases} 0, & \xi \leq \eta(1), \\ 1 - \mathfrak{J}^{-1}(\eta(0) - \xi), & \eta(1) < \xi < \eta(0), \\ 1, & \xi \geq \eta(0). \end{cases} \quad (\text{PH31})$$

Since $-\eta'(t) = f(t) \geq f_*$, its distribution function is globally L_* -Lipschitz. The exact grid distribution $H_q(\xi) = q^{-1} \#\{0 \leq j < q : \eta(j/q) \leq \xi\}$ satisfies $0 \leq \mathcal{H}(\xi) - H_q(\xi) \leq 1/q$, by counting the indices above the inverse cutoff. This includes both endpoints.

The absolute smoothing constant is

$$C_{\text{Gu}} = \mathbb{E}|G_{\text{Gu}}| = \int_0^\infty |\log x| e^{-x} dx = \gamma_E + 2E_1(1) \leq 1 + e^{-1}, \quad E_1(1) = \int_1^\infty e^{-x} \frac{dx}{x}. \quad (\text{PH32})$$

Here $\gamma_E = -\int_0^\infty \log x e^{-x} dx$ is Euler's constant. Split the integral at one and integrate the positive logarithmic part by parts to obtain the equality. For the elementary majorant, discard $e^{-x} \leq 1$ on $(0, 1)$, giving one, and use $\log x \leq x - 1$ on $(1, \infty)$, giving e^{-1} . Every stated heat error therefore has a finite explicit majorant without native arithmetic values.

Define $H_q^{\text{Gu}}(\xi) = \mathbb{E}\mathcal{H}(\xi - G_{\text{Gu}}/q)$. Matching each actual energy to its own cutoff centre and using the one-sided grid discrepancy gives

$$H_q^{\text{Gu}}(\xi - \delta_k) - 1/q \leq F_k(\xi) \leq H_q^{\text{Gu}}(\xi + \delta_k). \quad (\text{PH33})$$

Its convolution is L_* -Lipschitz and differs from $\mathcal{H}$ by at most $L_* C_{\text{Gu}}/q$. Consequently

$$\sup_\xi |F_k(\xi) - \mathcal{H}(\xi)| \leq L_* \delta_k + \frac{1}{q} + \frac{L_* C_{\text{Gu}}}{q} =: \varepsilon_k^{\text{heat}}. \quad (\text{PH34})$$

This replaces the supplied $\log q/q$ smoothing cost by C_{Gu}/q . The supplied bound itself is valid: use the Gumbel tails at $\pm \log q$, which are e^{-q} on the left and at most $1/q$ on the right, in the same shifted distribution argument. This gives $L_*(\delta_k + \log q/q) + 2/q + e^{-q}$.

The primary original measured observable retains its one-sided defect:

$$\mathcal{H}(\xi) - \varepsilon_k^{\text{heat}} \leq M_k(\xi) \leq \mathcal{H}(\xi) + \varepsilon_k^{\text{heat}} + D_k^{\text{ang}}(\xi) \leq \mathcal{H}(\xi) + \varepsilon_k^{\text{heat}} + \alpha_k. \quad (\text{PH35})$$

Thus the stated uniform limit follows from the proved $\delta_k \rightarrow 0$ and $\alpha_k \rightarrow 0$. No current insertion occurs in this identity or its limiting argument.

6. The probability observable and its exact map to the primary heat

An additional probability observable has an exact affine map to M_k ; it does not replace it. On $\alpha_k < 1$, subtract the actual observed zero trace PH25 and divide by the actual remaining mass:

$$P_k^{\text{obs}}(\xi) = \frac{\sum_j [\text{Tr}_B \mathcal{Z}_{N_j}(\tau_k(\xi)) - (q - m - \Delta + \vartheta_{N_j}^\Gamma)]}{q\Delta - \sum_j \vartheta_{N_j}^\Gamma} = \frac{M_k(\xi) - \alpha_k}{1 - \alpha_k}. \quad (\text{PH36})$$

On the native family $\Delta > m$ and $\vartheta_N^\Gamma \leq m$, so this guard holds. For each full energy projector,

$$w_{\lambda,N} = \text{Tr}(J_{B,N}^* G_N P_{\lambda,N} J_{B,N}) = \text{rank } P_{\lambda,N} - \text{Tr}(J_{K,N}^* G_N P_{\lambda,N} J_{K,N}) \geq 0.$$

These positive observed weights sum to $\Delta - \vartheta_N^\Gamma$. Hence P_k^{obs} is the probability distribution of the actual weighted positive exponents convolved with the same Gumbel law, with every repeated-energy projector retained.

For $\alpha_k > 0$, form R_k^{rem} from the complementary nonnegative kernel projector weights, normalized by $q\Delta\alpha_k$, and the same convolution. Then exactly

$$F_k = (1 - \alpha_k)P_k^{\text{obs}} + \alpha_k R_k^{\text{rem}}, \quad M_k = \alpha_k + (1 - \alpha_k)P_k^{\text{obs}}, \quad D_k^{\text{ang}} = \alpha_k(1 - R_k^{\text{rem}}). \quad (\text{PH37})$$

In particular $\sup |P_k^{\text{obs}} - F_k| \leq \alpha_k$. If $\alpha_k = 0$, the primary, full and observed probability functions coincide and no removed distribution is required.

A stronger global transport estimate follows from an explicit coupling. Let U be uniform on $[0, 1]$, take $j = \lfloor qU \rfloor$, and choose a positive energy index uniformly at that cutoff. Since η decreases, the mean absolute distance between its grid centre and $\eta(U)$ is

$$R_q^\eta = \frac{1}{q} \sum_{j=0}^{q-1} \eta(j/q) - \int_0^1 \eta(t) dt \in [0, \mathfrak{J}(1)/q].$$

The actual energy adds at most δ_k to this displacement; the independent Gumbel adds mean absolute displacement C_{Gu}/q . Thus, for W_1 defined as the infimum of mean absolute displacement over all couplings,

$$W_1(dF_k, d\mathcal{H}) \leq \delta_k + R_q^\eta + C_{\text{Gu}}/q \leq \delta_k + \frac{\mathfrak{J}(1) + C_{\text{Gu}}}{q}. \quad (\text{PH38})$$

For any two coupled variables, the integral of the absolute difference of their half-line indicators is their absolute distance. Tonelli therefore bounds the global integral of the difference of their distribution functions by the same right-hand side. No pointwise density convergence is needed.

All unsmoothed exponents lie in $[\eta(1) - \delta_k, \eta(0) + \delta_k]$, of length $\mathfrak{J}(1) + 2\delta_k$. In the observed/removed decomposition, couple the common observed mass identically and the remaining mass inside this interval. Its cost is at most $\alpha_k[\mathfrak{J}(1) + 2\delta_k]$. Convolution with a common Gumbel variable preserves that bound, so

$$W_1(dP_k^{\text{obs}}, d\mathcal{H}) \leq \delta_k + R_q^\eta + C_{\text{Gu}}/q + \alpha_k[\mathfrak{J}(1) + 2\delta_k]. \quad (\text{PH39})$$

The primary M_k has left limit α_k ; when $\alpha_k > 0$ its global L^1 distance from $\mathcal{H}$ is infinite. PH36–39 specify the exact probability space and affine map that remove this defect while PH27–35 preserve the original observable.

The physical heat smoothing bias is exact:

$$\int_{\mathbb{R}} \xi dF_k(\xi) = \frac{1}{q\Delta} \sum_{j,a} u_{a,j} + \frac{\gamma_E}{q}. \quad (\text{PH40})$$

This follows from $\mathbb{E}G_{\text{Gu}} = \gamma_E$. The analogous identity uses the actual observed positive weights for P_k^{obs} . The declared physical heat time has not been shifted to absorb this correction.

Inside the limiting transition interval,

$$\mathcal{H}'(\xi) = \frac{1}{f(t_\xi)}, \quad t_\xi = \mathfrak{J}^{-1}(\eta(0) - \xi); \quad (\text{PH41})$$

outside it the density is zero. The endpoints have no atoms because their inverse images under the continuous strictly decreasing η have Lebesgue measure zero. The finite heat densities are nonnegative mixtures of $qe^{-q(\xi-u)} \exp[-e^{-q(\xi-u)}]$. PH38–39 prove convergence of their measures and bounded Lipschitz tests. They do not assert pointwise or strong convergence of density derivatives, which is not implied by a grid spacing and a smoothing width both of order $1/q$.

7. Scope and exact verification

CP37–51 and the supplied additional heat assertions are valid with their stated finite errors. PH9 supplies a finite full-image guard and retains both absolute source factors ℓ_k, u_k ; only their ratio cancels in cutoff comparisons. PH34 sharpens the heat error, PH28 retains its exact original angle defect, and PH36–40 give the associated probability and transport maps.

The complete coefficient, quotient and image comparisons used are CK3, CK5 and CK9–11, read at lines23–145 of the published source. The companion power derivation owns CP9/27/33. Original-metric Schur identities are also proved in 021 MR1–29 and rederived where used here. No positive heat expression is identified with the separate current $i(Q_{B,N}M_{B,N} - M_{B,N}^*Q_{B,N})$ or with its marked-class sign.

The human formula sources retained in PH11 and PH29 are R. A. Askey and R. Roy, Gamma product, DLMF 5.8.3, and T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, Legendre generating function, 18.12.11, Meixner–Pollaczek recurrence, 18.22.8 and generating function, 18.23.7. The original formula TeX files were read in full. J. L. W. V. Jensen’s zero-count formula is credited in the retained AS proof; its finite disk use is proved there, and no new reading of Jensen’s historical article is claimed here. The exponential-to-Gumbel probability change and every smoothing and transport bound used in PH30–41 are derived directly above. Naming the classical Gumbel distribution does not claim that the distribution itself is new.

The standalone checker is `check_phase_heat.py`. Its exact rational and complex-algebraic fixtures verify the full polynomial map, every tested relation column, full metric sections, conductor compression, zero-energy factors, scalar derivative identities, heat defects, probability map, tail constants and finite cutoff couplings. These auxiliary checks do not substitute chosen fixture values for the original period data.